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(54) **MULTIPLE LOCATION LOAD CONTROL SYSTEM**

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G11C 5/00 (2006.01)
(Continued)

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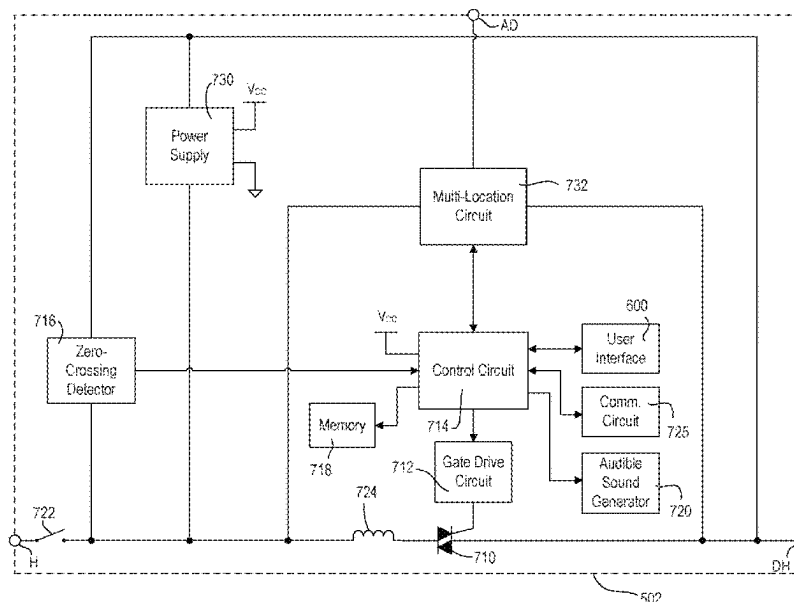
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(57) **ABSTRACT**

A load control device may include a semiconductor switch, a control circuit, and first and second terminals adapted to be coupled to a remote device. The load control device may include a first switching circuit coupled to the second terminal, and a second switching circuit coupled between the first terminal and the second terminal. The control circuit may be configured to render the first switching circuit conductive to conduct a charging current from an AC power source to a power supply of the remote device during a first time period of a half-cycle of the AC power source, and further configured to render the first and second switching circuits conductive and non-conductive to communicate with the remote device via the second terminal during a second time period of the half-cycle of the AC power source.

18 Claims, 21 Drawing Sheets



Related U.S. Application Data

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continuation of application No. 17/402,809, filed on Aug. 16, 2021, now Pat. No. 11,558,939, which is a continuation of application No. 16/811,226, filed on Mar. 6, 2020, now Pat. No. 11,094,353, which is a continuation of application No. 16/179,317, filed on Nov. 2, 2018, now Pat. No. 10,593,373, which is a continuation of application No. 15/611,882, filed on Jun. 2, 2017, now Pat. No. 10,129,948, which is a continuation of application No. 14/720,701, filed on May 22, 2015, now Pat. No. 9,699,863.

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G11C 11/406 (2006.01)
G11C 29/12 (2006.01)
H05B 39/04 (2006.01)
H05B 47/10 (2020.01)
H05B 47/165 (2020.01)
H05B 47/17 (2020.01)
H05B 47/185 (2020.01)

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(58) Field of Classification Search

CPC G11C 5/00; G11C 5/005; G11C 5/025; G11C 7/04; G11C 7/24; G11C 11/40626; G11C 29/12; G11C 11/406; G11C 5/02
 See application file for complete search history.

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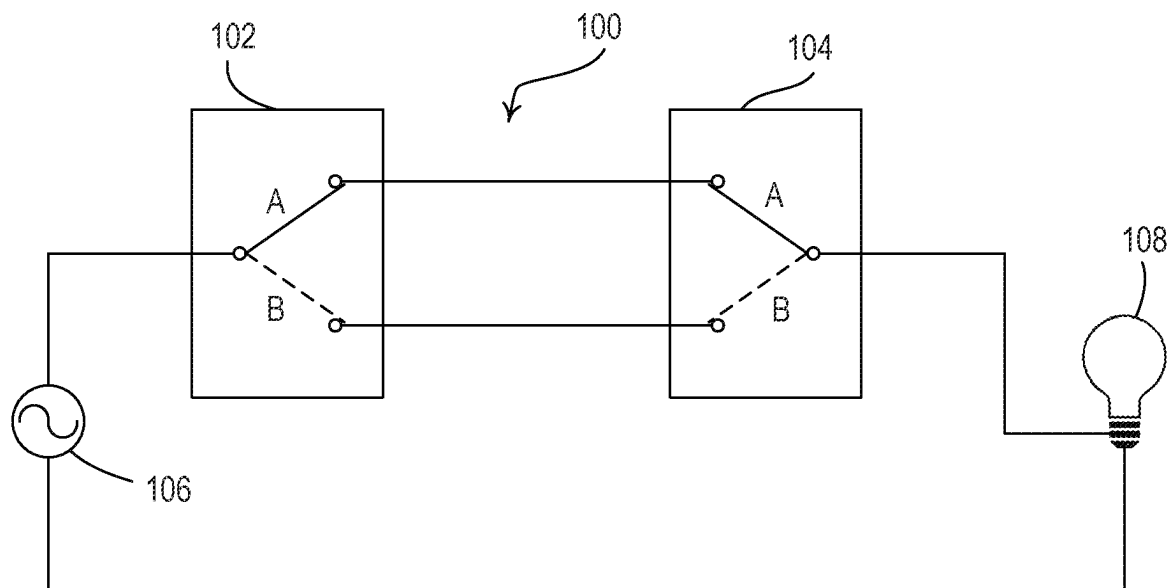


Fig. 1A
Prior Art

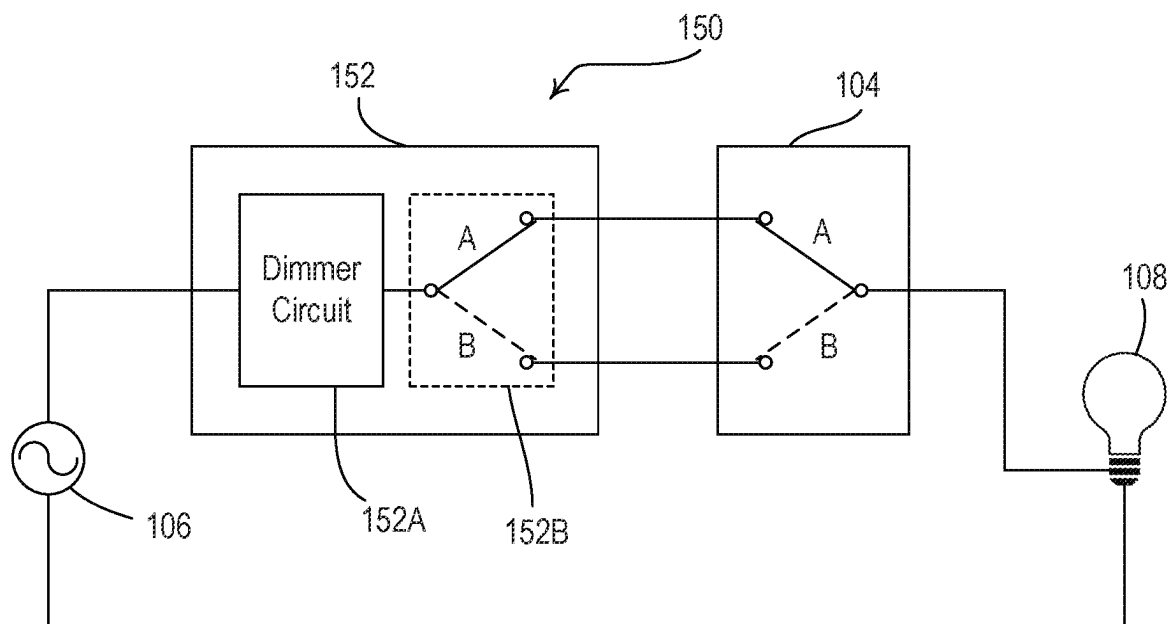


Fig. 1B
Prior Art

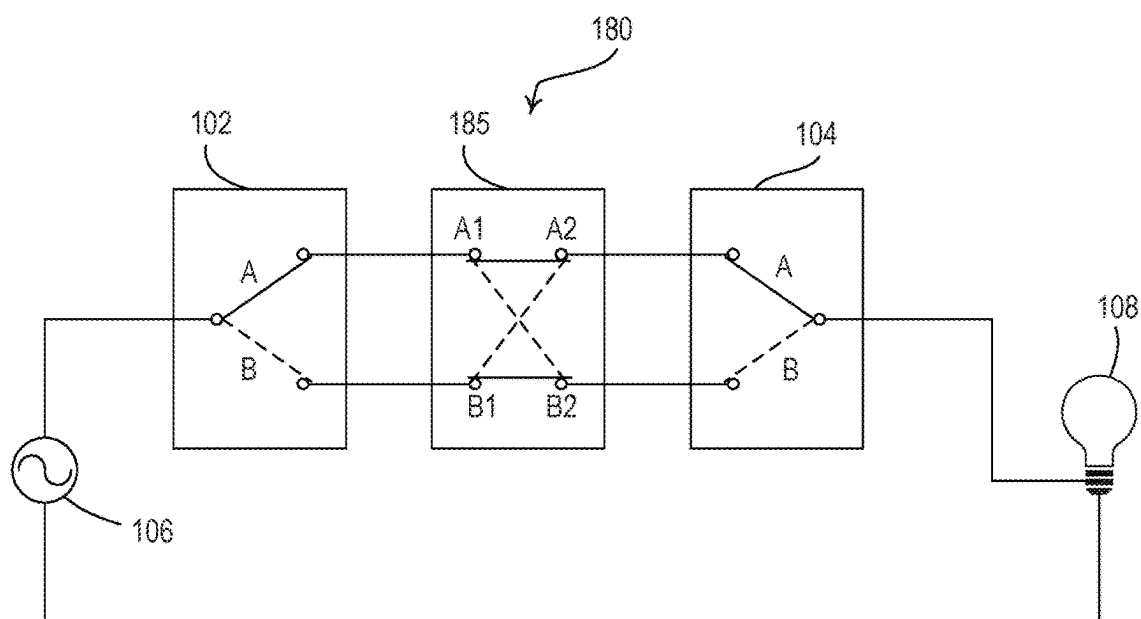


Fig. 1C
Prior Art

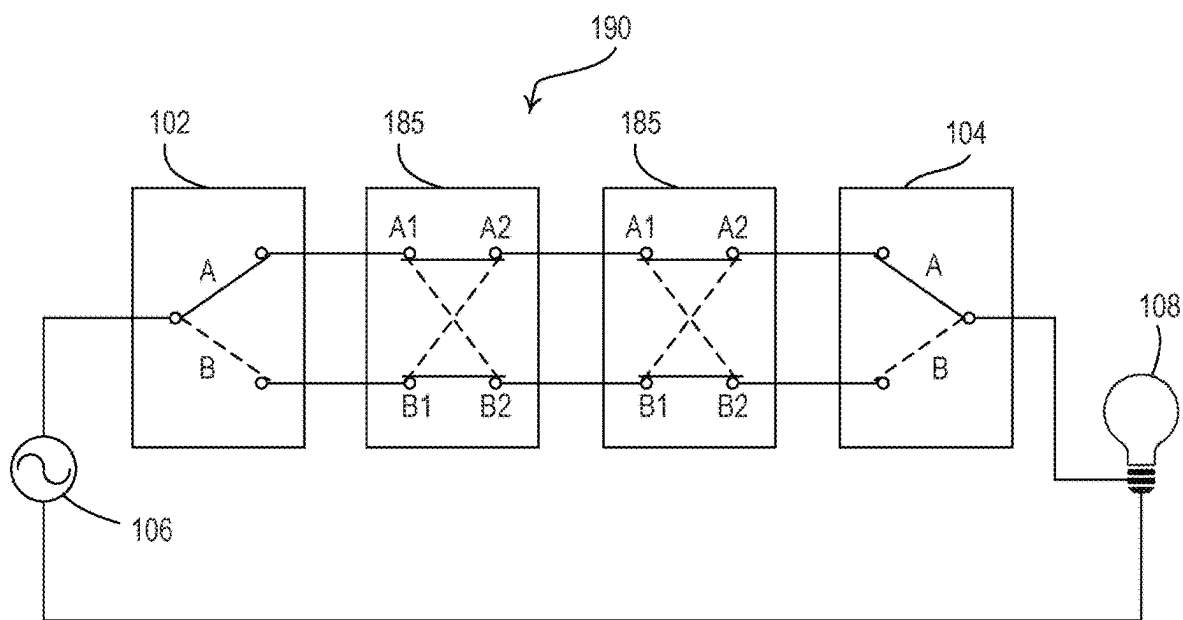


Fig. 1D
Prior Art

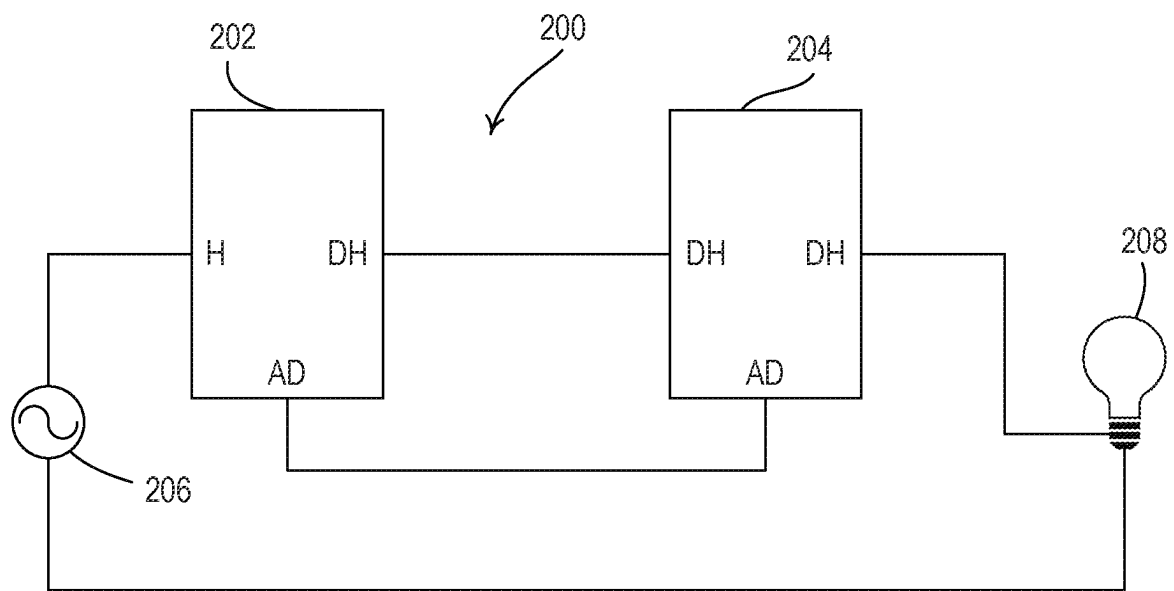


Fig. 2
Prior Art

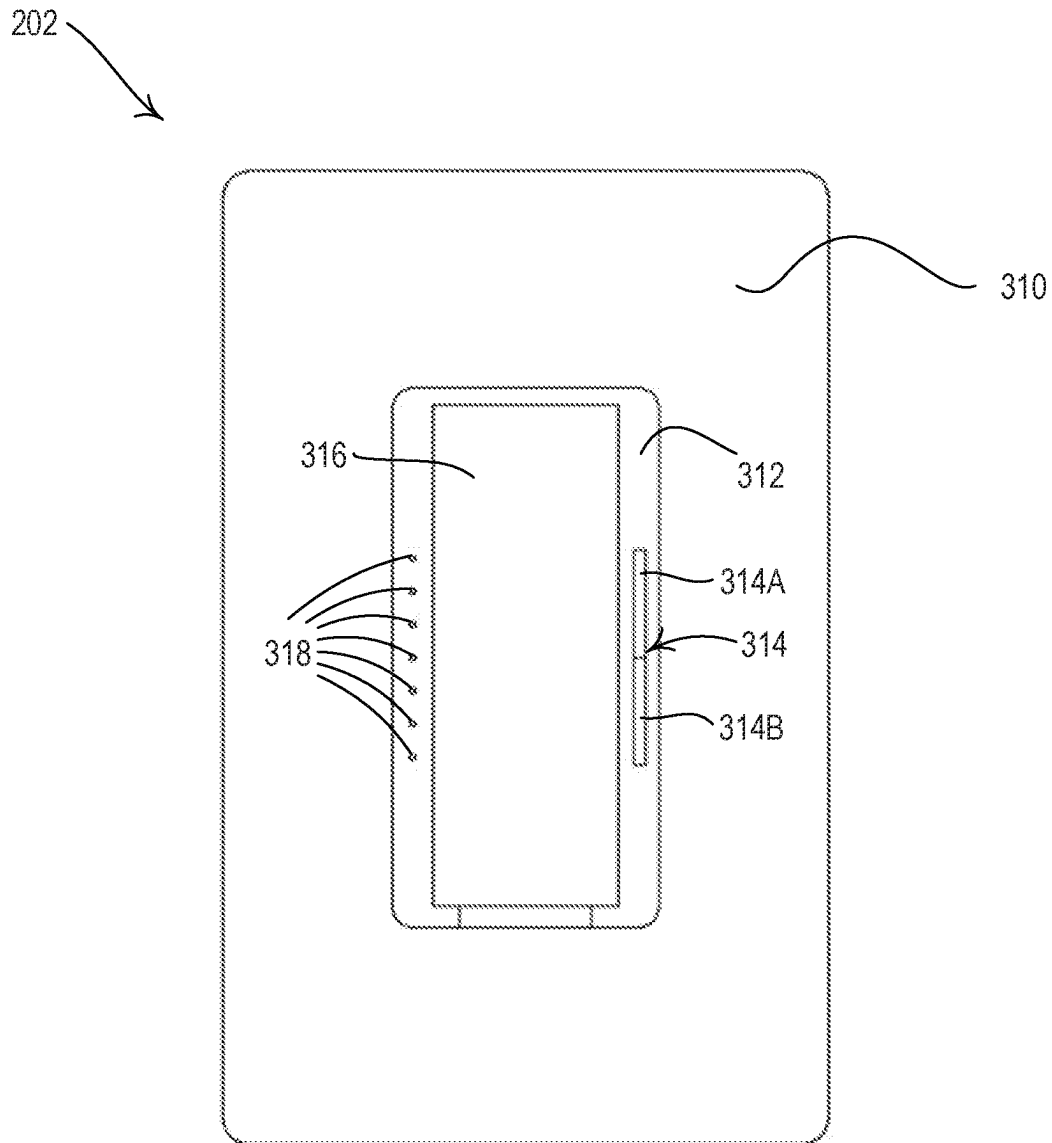


Fig. 3
Prior Art

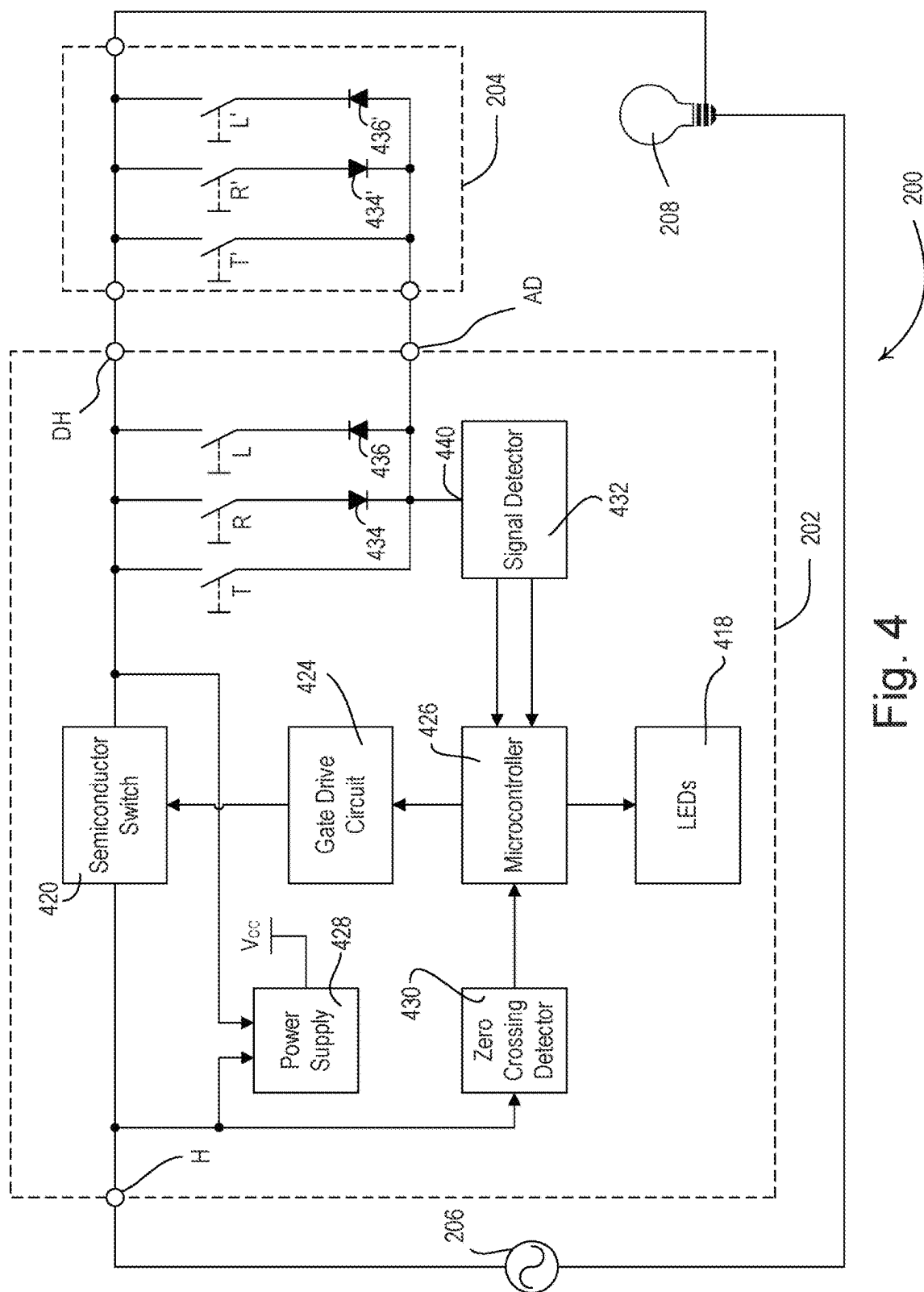


Fig. 4
Prior Art

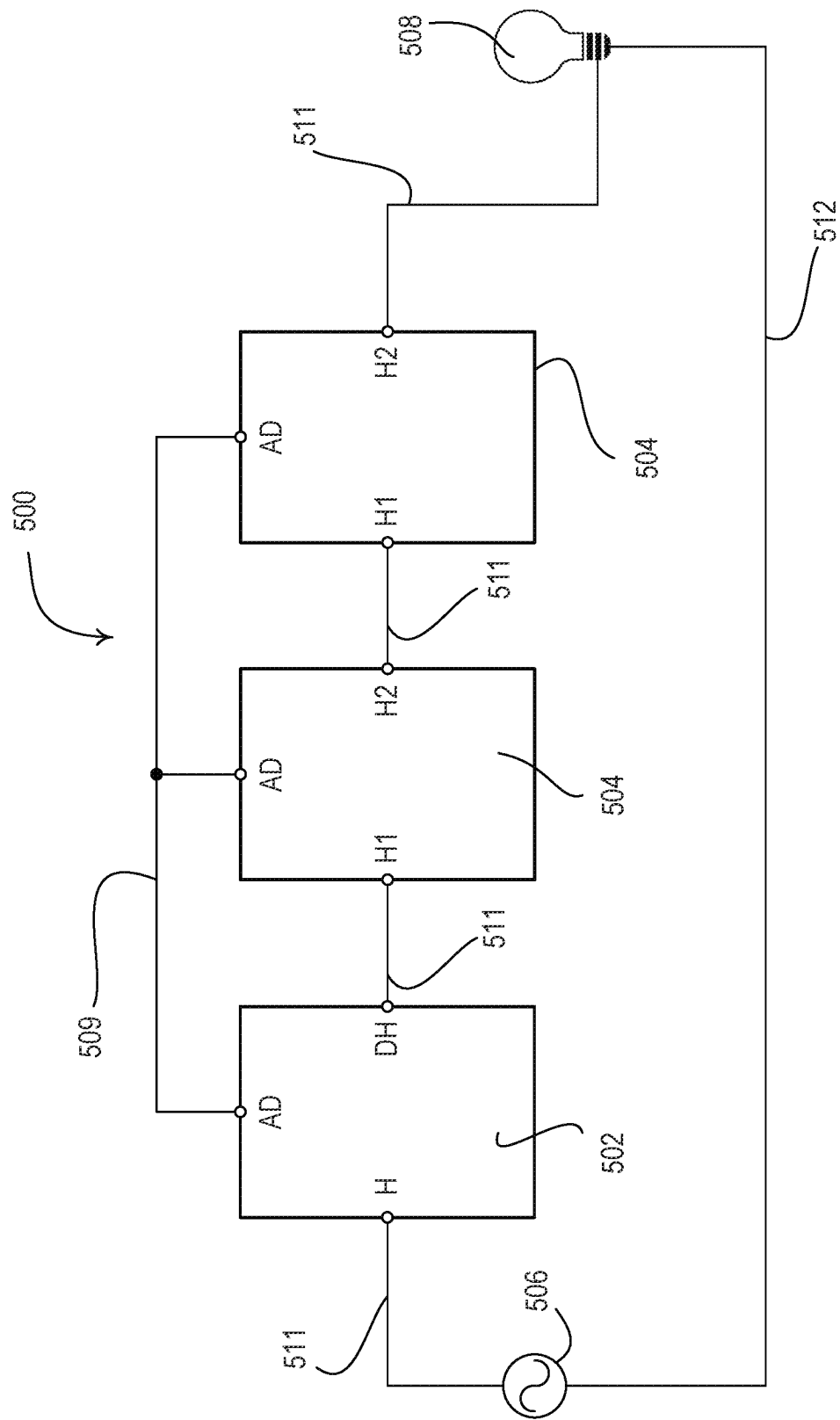


Fig. 5A

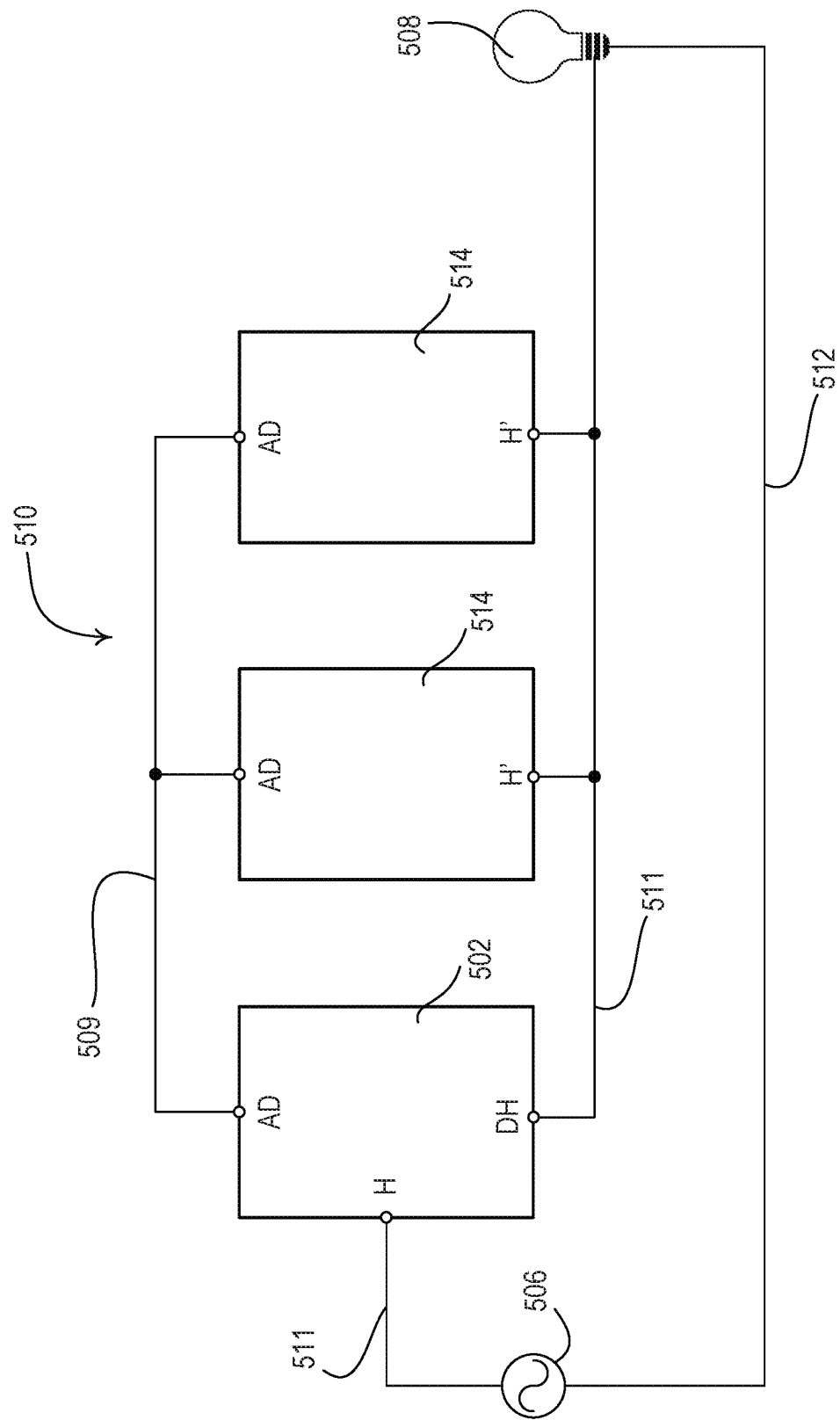


Fig. 5B

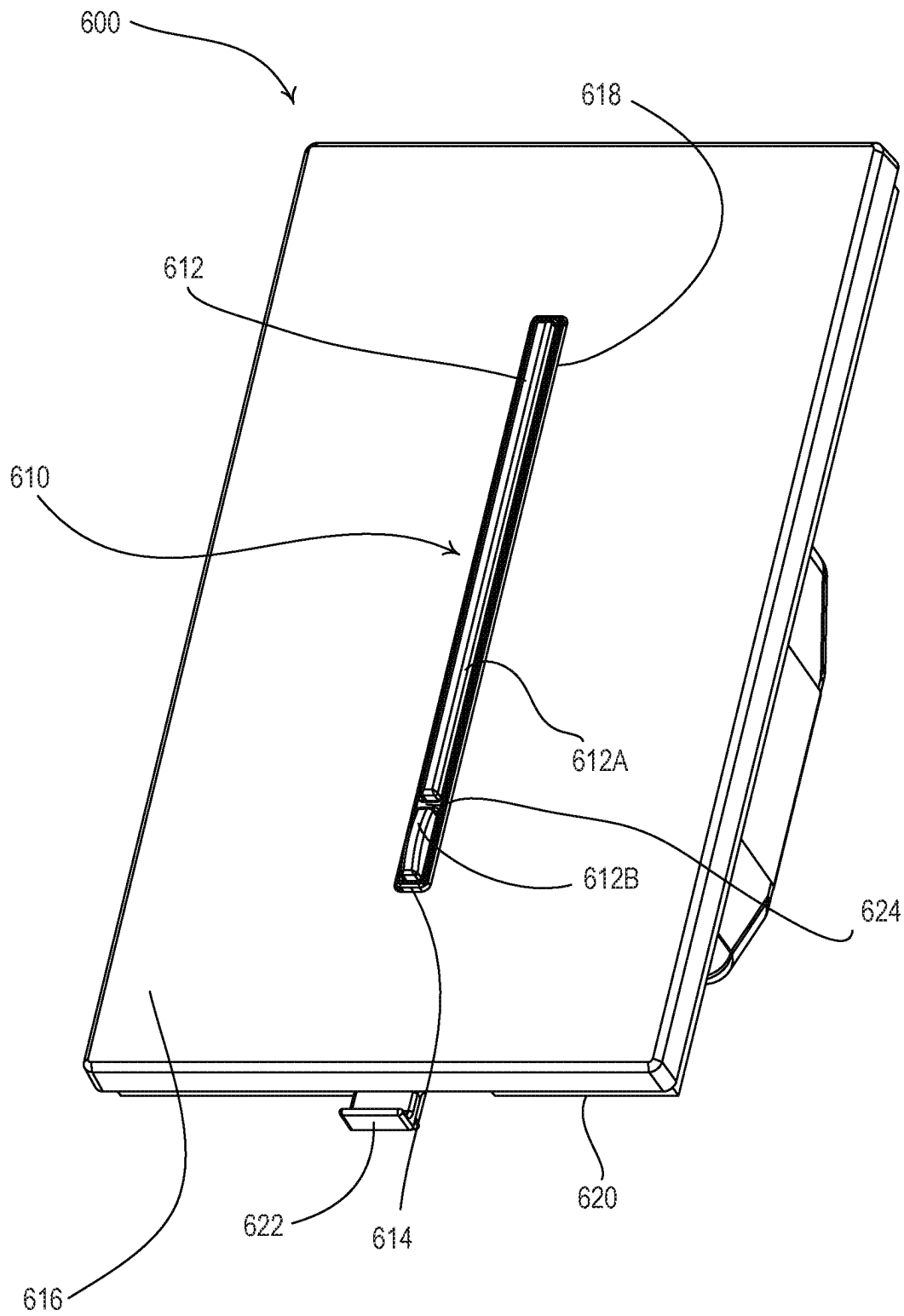


Fig. 6

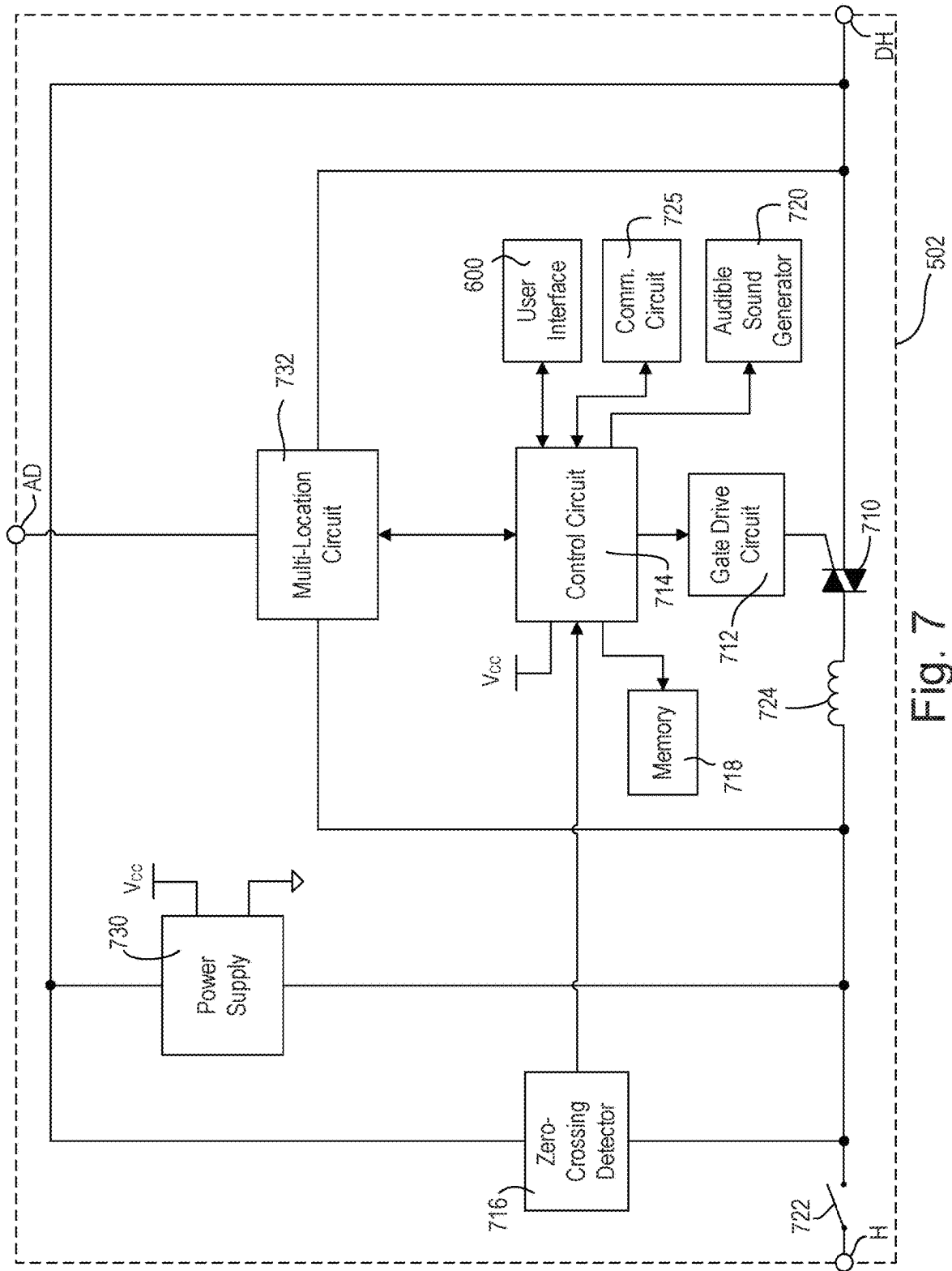


Fig. 7

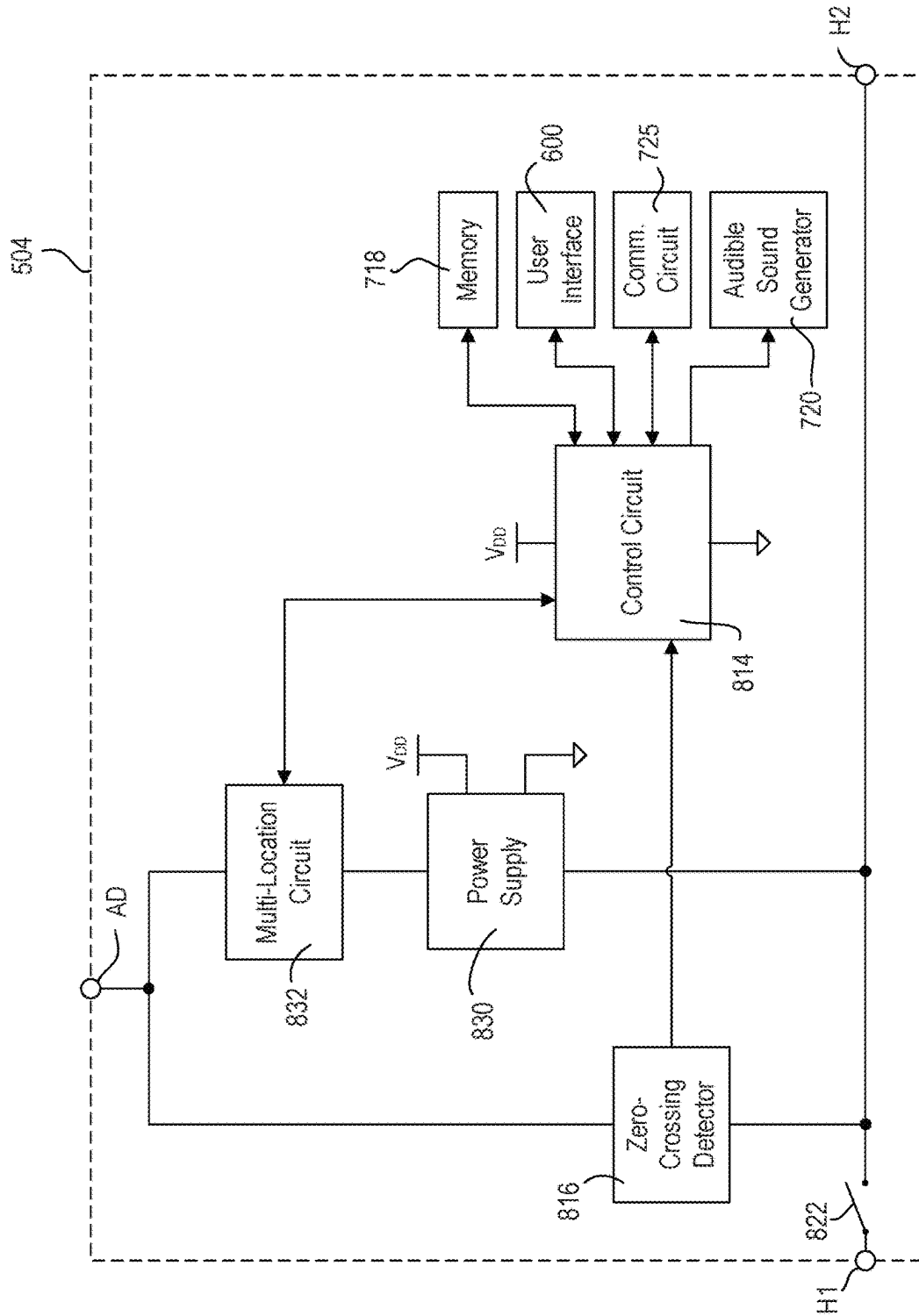


Fig. 8A

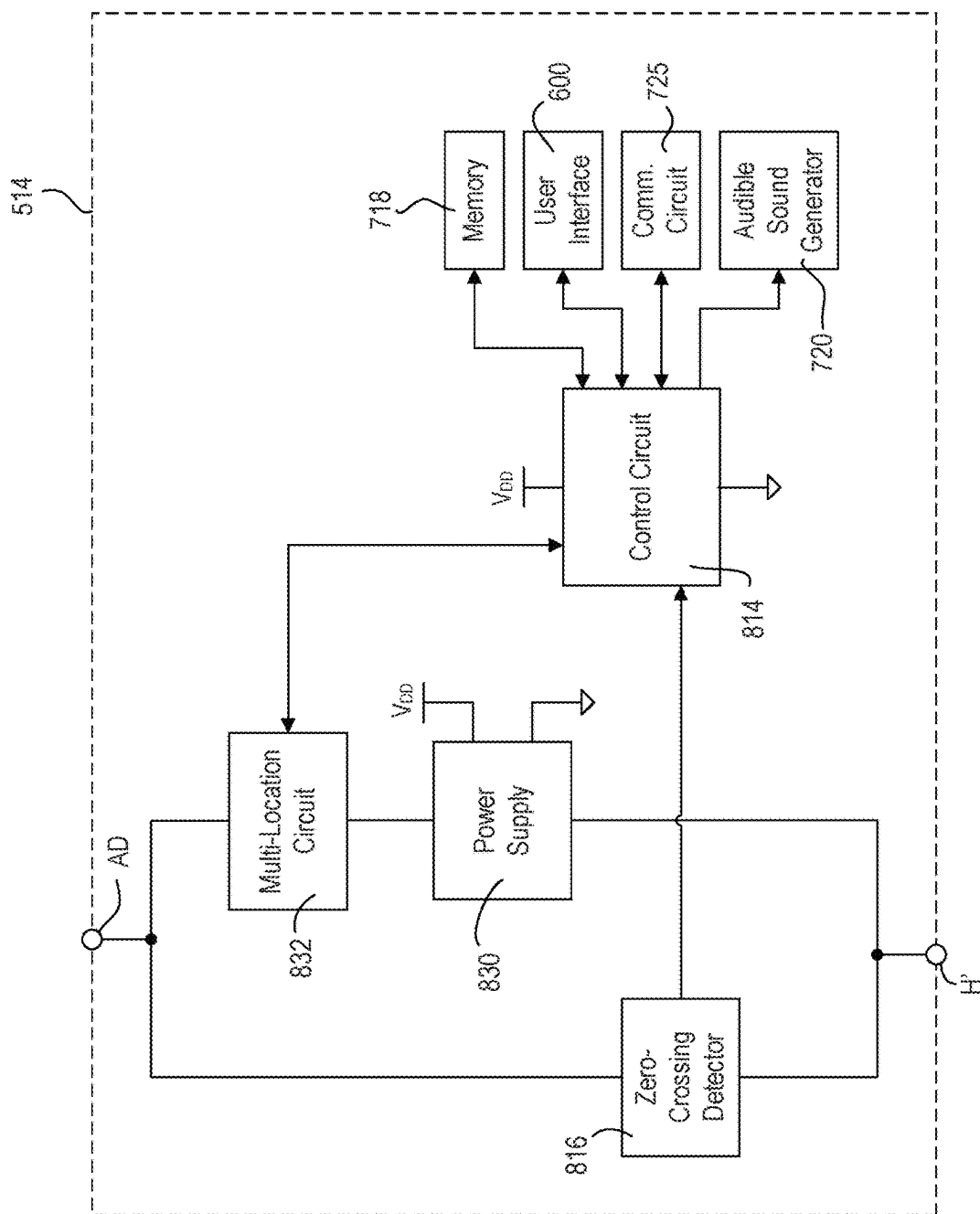


Fig. 8B

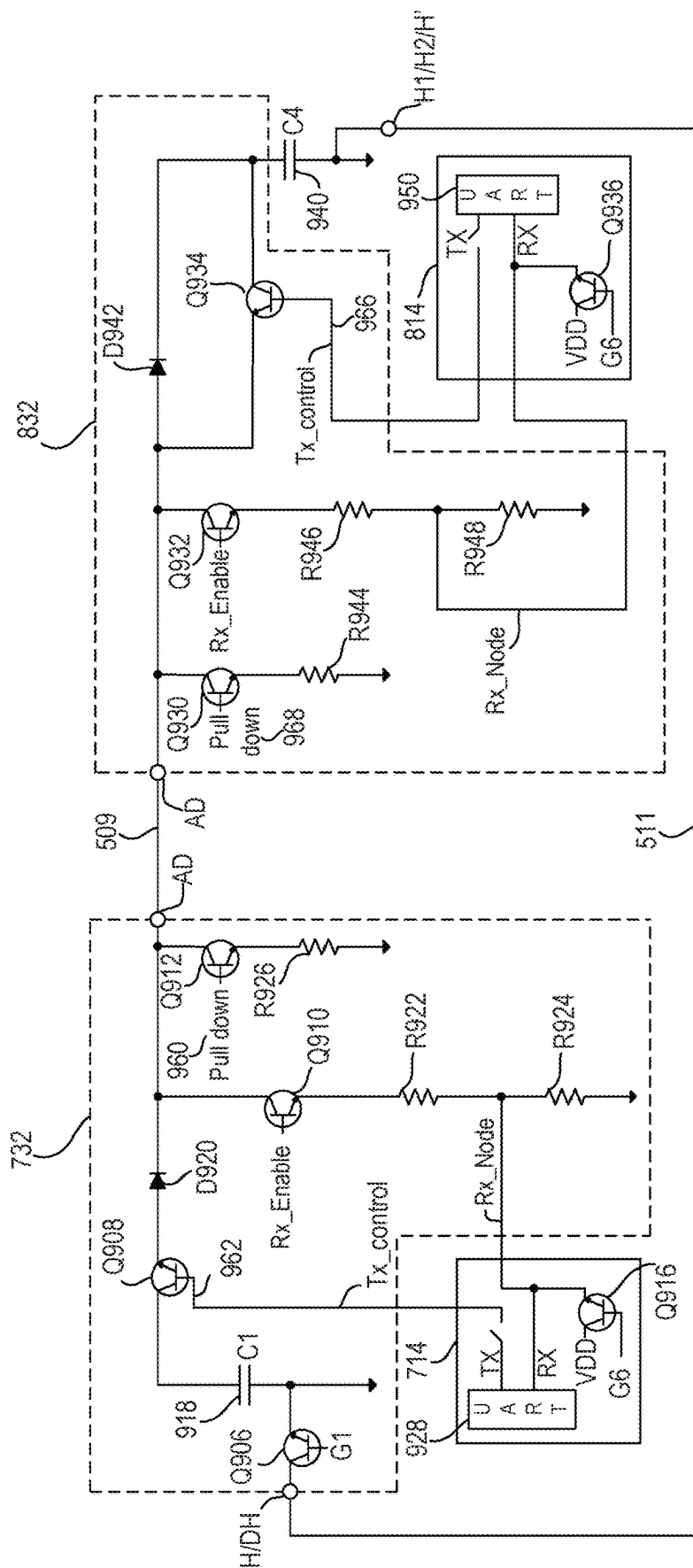


Fig. 9

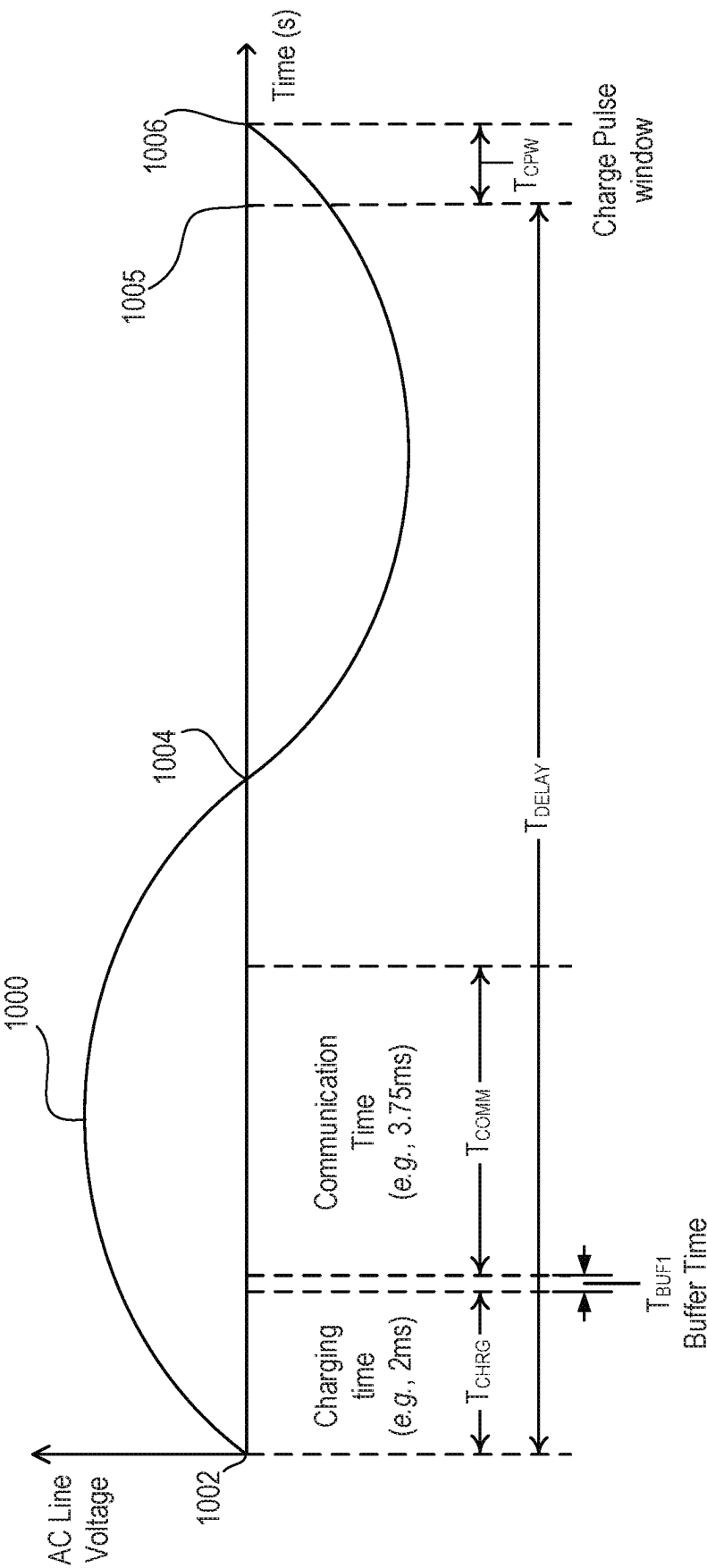


Fig. 10

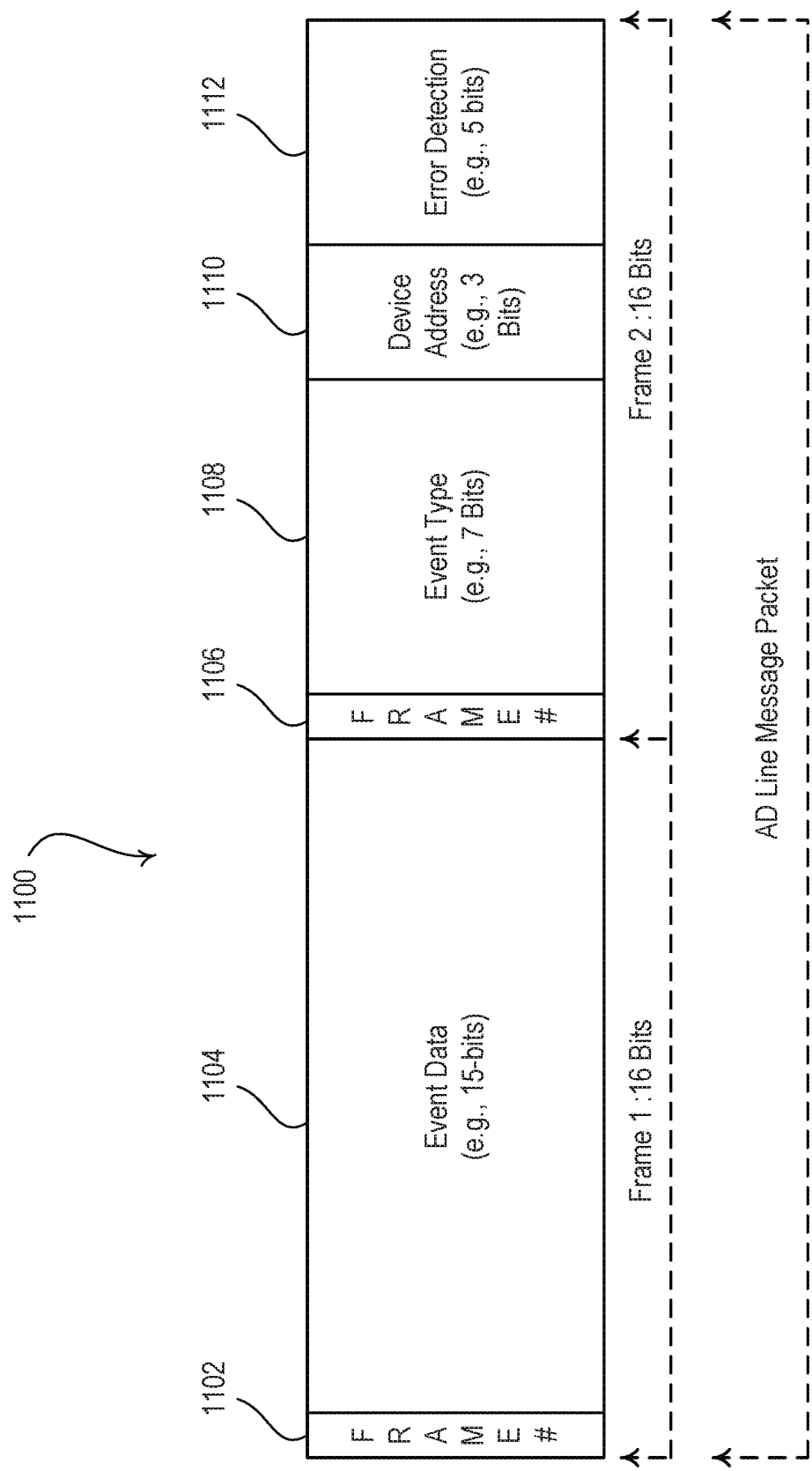


Fig. 11

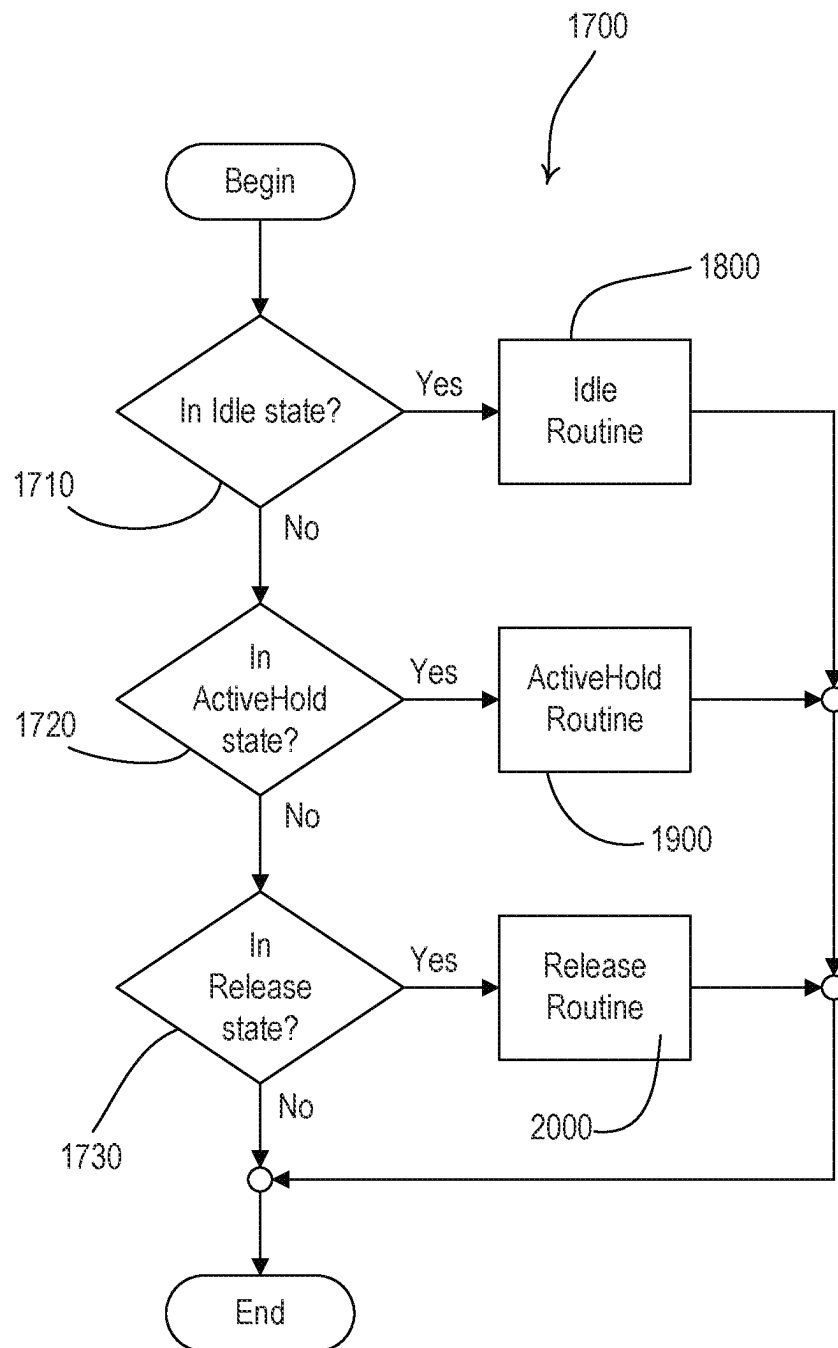


Fig. 12

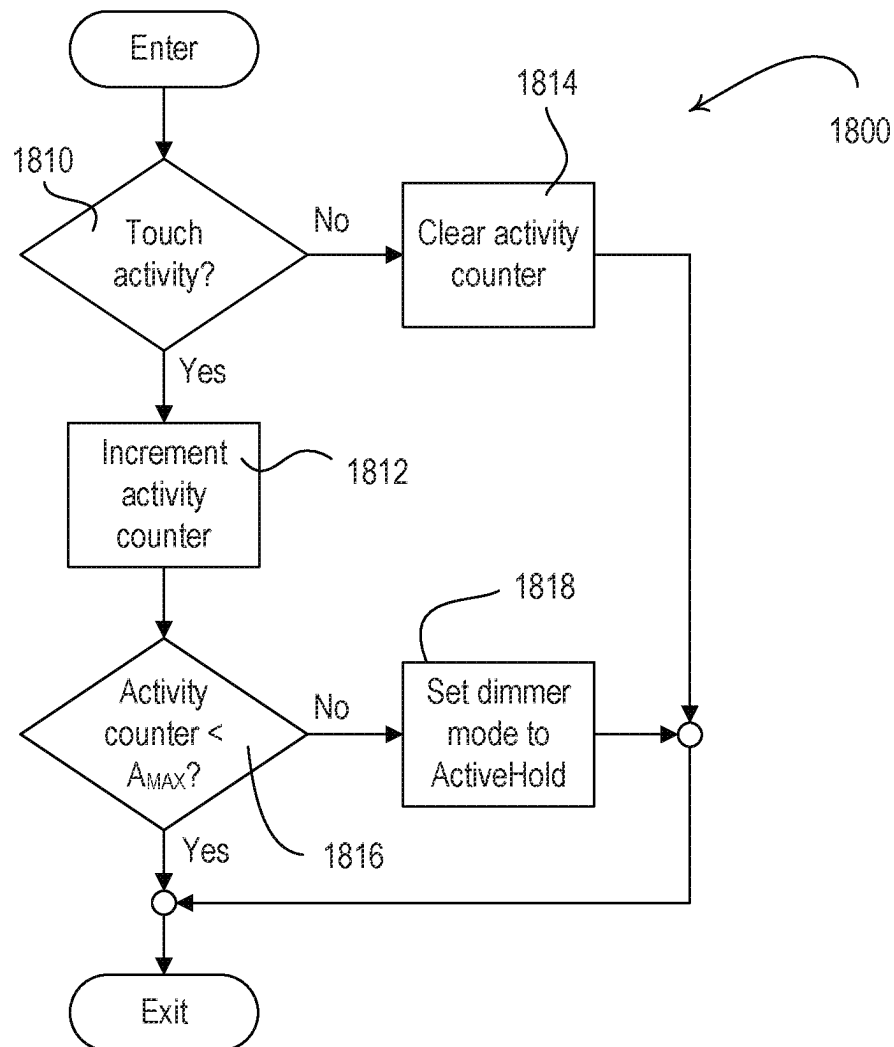


Fig. 13

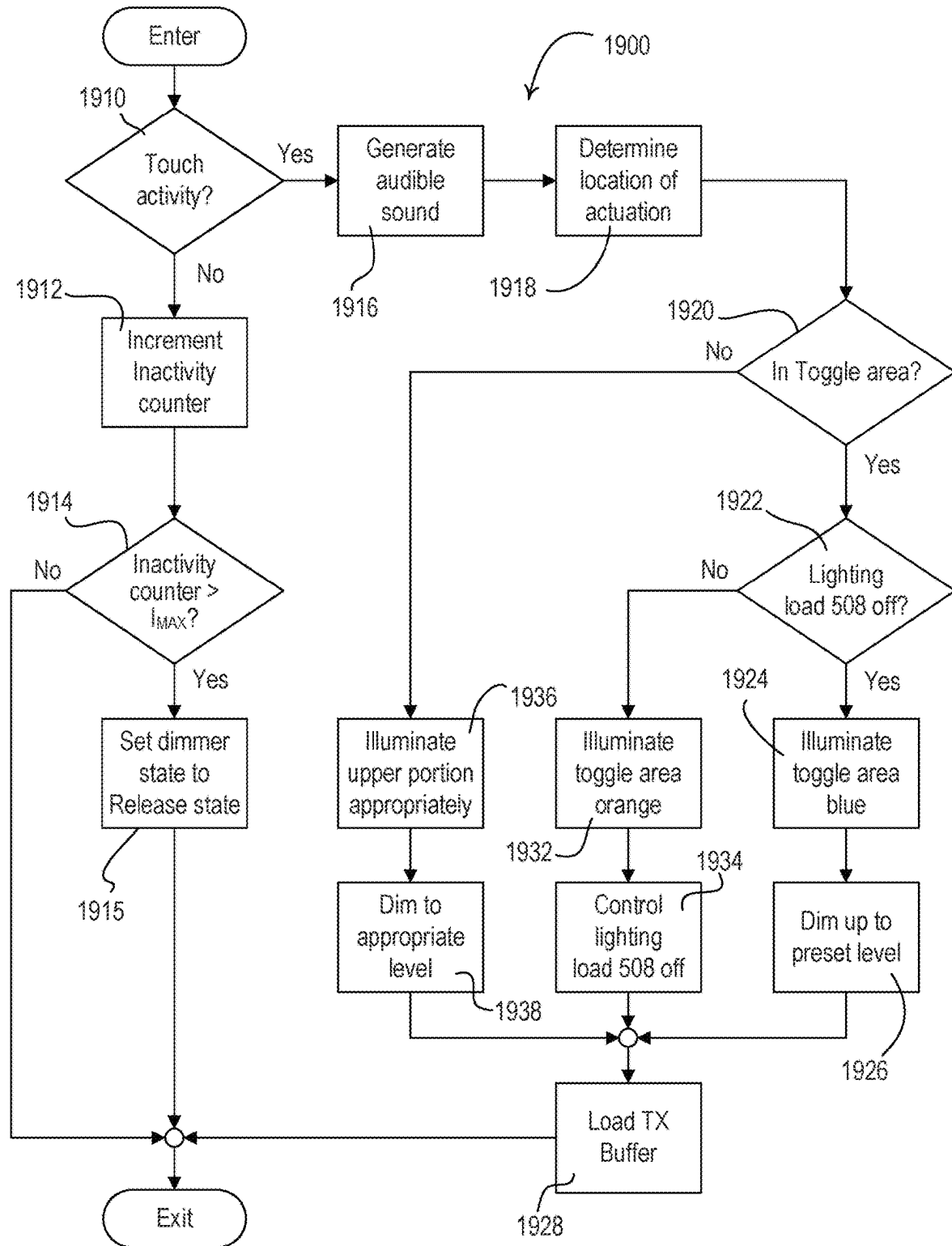


Fig. 14

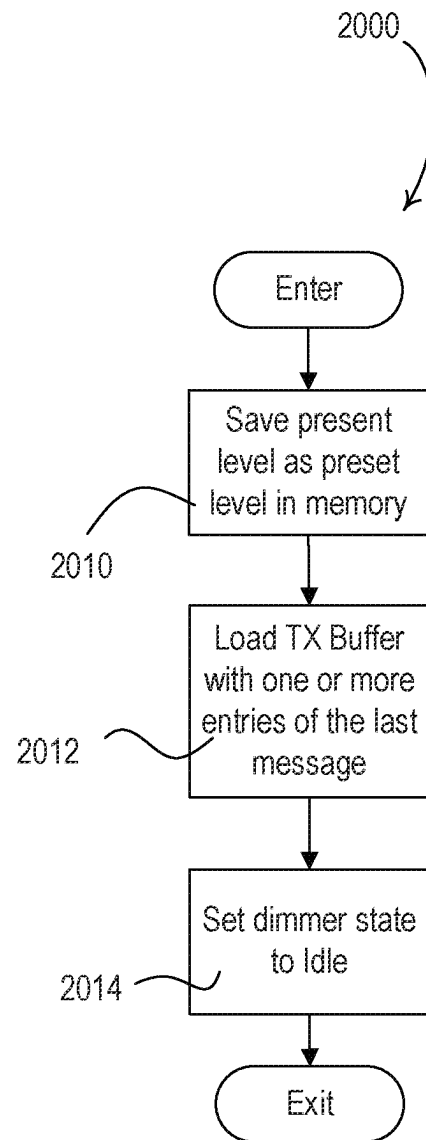


Fig. 15

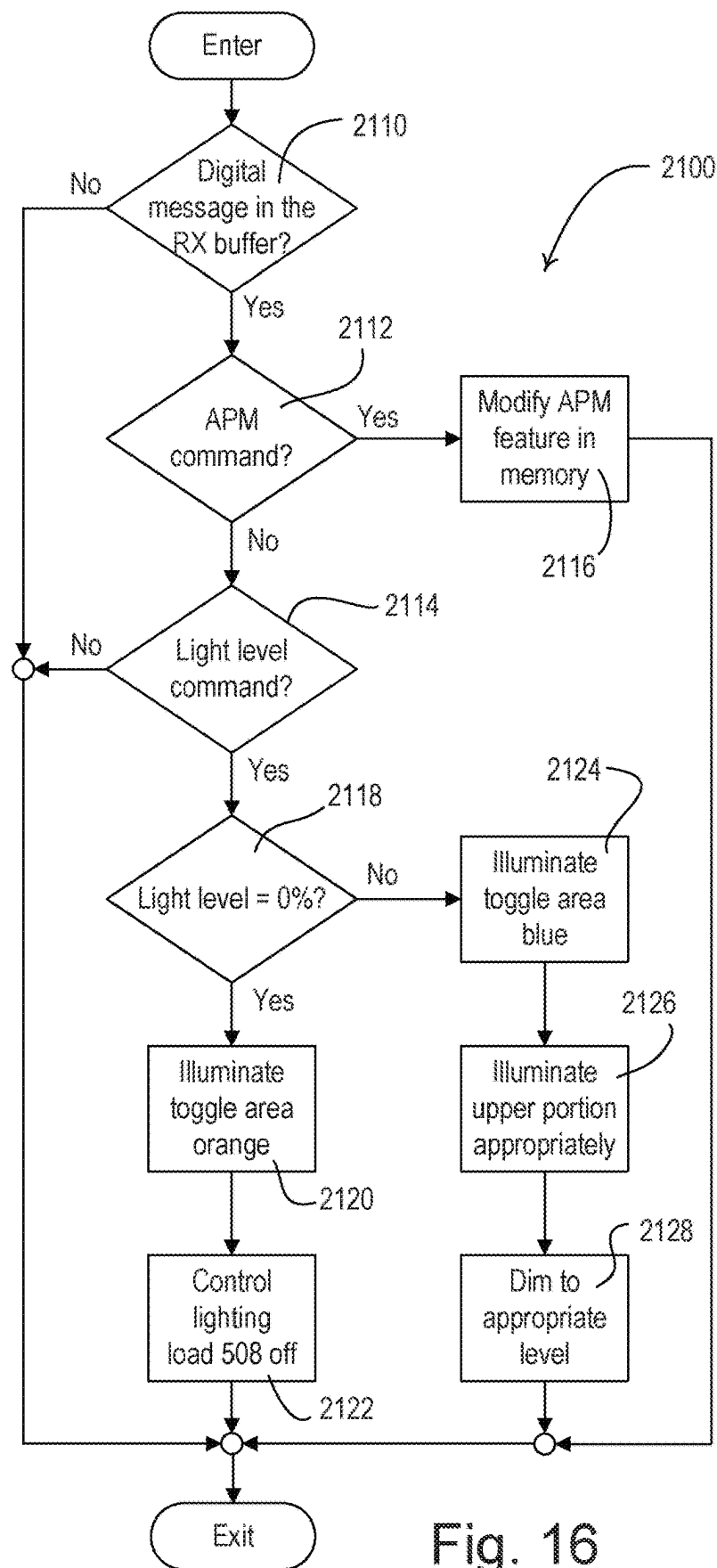
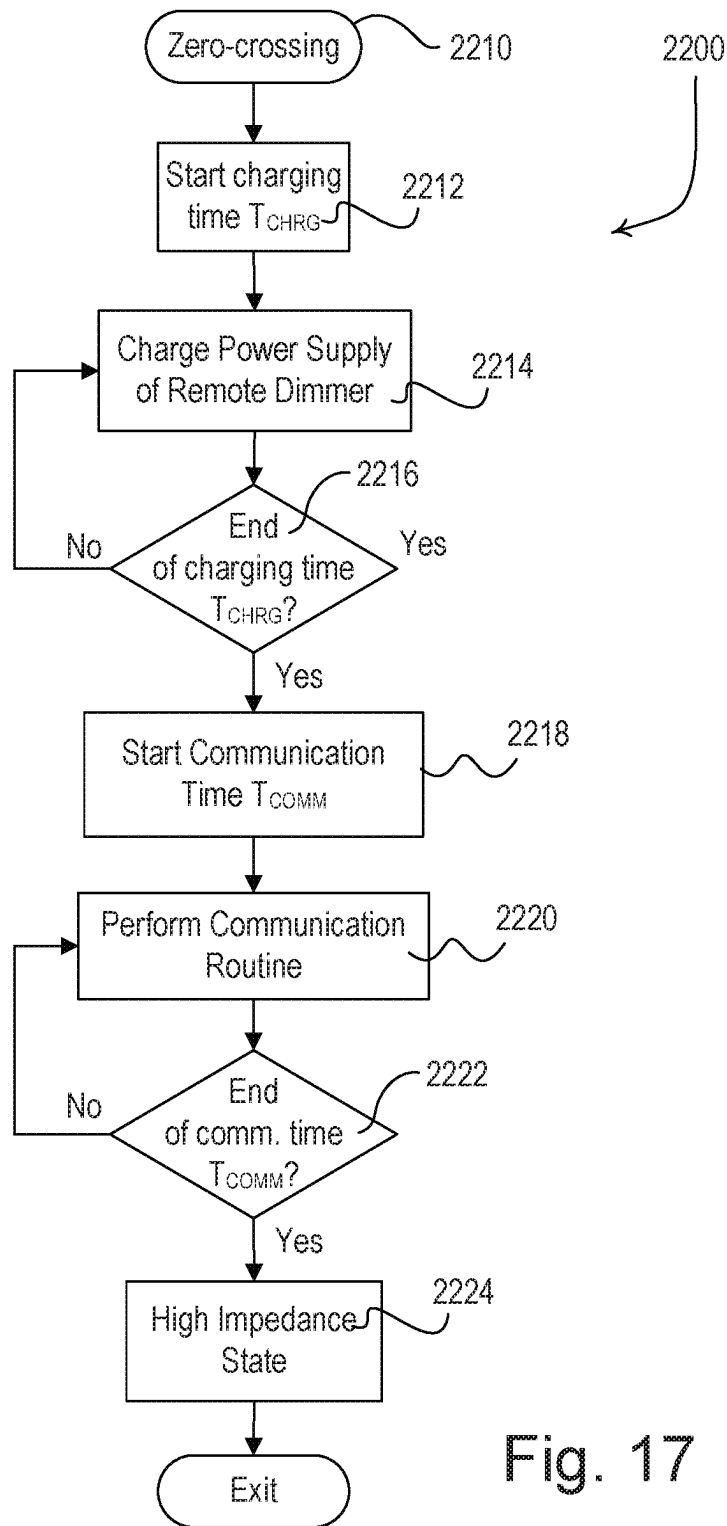


Fig. 16



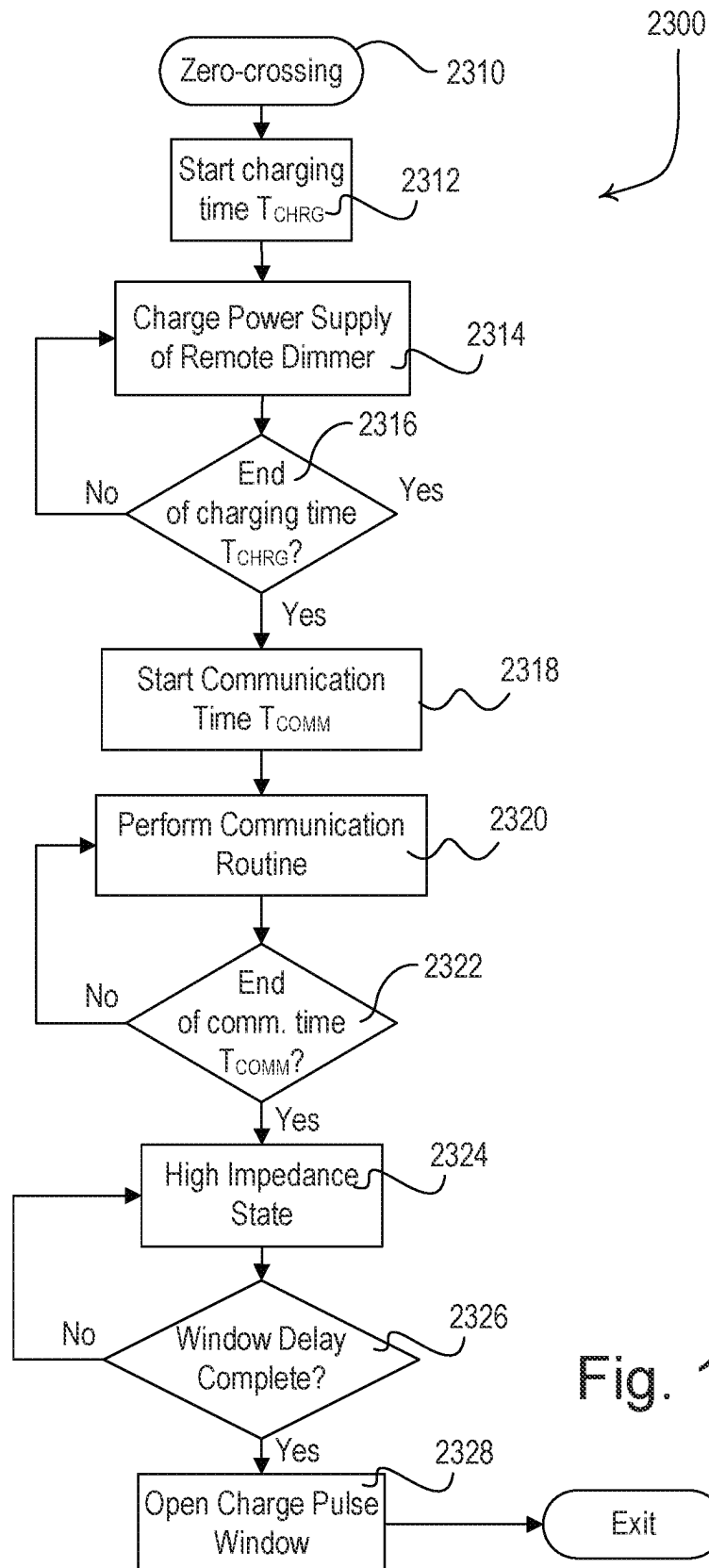


Fig. 18

MULTIPLE LOCATION LOAD CONTROL SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/154,907, filed Jan. 16, 2023; which is a continuation of U.S. patent application Ser. No. 17/402,809, filed Aug. 16, 2021, now U.S. Pat. No. 11,558,939, issued Jan. 17, 2023; which is a continuation of U.S. patent application Ser. No. 16/811,226, filed Mar. 6, 2020, now U.S. Pat. No. 11,094,353, issued Aug. 17, 2021; which is a continuation of U.S. patent application Ser. No. 16/179,317, filed Nov. 2, 2018, now U.S. Pat. No. 10,593,373 issued Mar. 17, 2020; which is a continuation of U.S. patent application Ser. No. 15/611,882, filed Jun. 2, 2017, now U.S. Pat. No. 10,129,948, issued on Nov. 13, 2018; which is a continuation of U.S. patent application Ser. No. 14/720,701, filed May 22, 2015, now U.S. Pat. No. 9,699,863, issued on Jul. 4, 2017; which claims the benefit of U.S. Provisional Application No. 62/005,922, filed May 30, 2014, the entire contents of which are incorporated by reference herein.

BACKGROUND

Three-way and four-way switch systems for use in controlling electrical loads, such as lighting loads, are known in the art. Typically, the switches are coupled together in series electrical connection between an alternating-current (AC) power source and the lighting load. The switches are subjected to an AC source voltage and carry full load current between the AC power source and the lighting load, as opposed to low-voltage switch systems that operate at low voltage and low current, and communicate digital commands (usually low-voltage logic levels) to a remote controller that controls the level of AC power delivered to the load in response to the commands. Thus, as used herein, the terms “three-way switch”, “three-way system”, “four-way switch”, and “four-way system” mean such switches and systems that are subjected to the AC source voltage and carry the full load current.

A three-way switch derives its name from the fact that it has three terminals and is more commonly known as a single-pole double-throw (SPDT) switch, but will be referred to herein as a “three-way switch”. Note that in some countries a three-way switch as described above is known as a “two-way switch”.

A four-way switch is a double-pole double-throw (DPDT) switch that is wired internally for polarity-reversal applications. A four-way switch is commonly called an intermediate switch, but will be referred to herein as a “four-way switch”.

In a typical, prior art three-way switch system, two three-way switches control a single lighting load, and each switch is fully operable to independently control the load, irrespective of the status of the other switch. In such a three-way switch system, one three-way switch must be wired at the AC power source side of the system (sometimes called “line side”), and the other three-way switch must be wired at the lighting load side of the system.

FIG. 1A shows a standard three-way switch system **100**, which includes two three-way switches **102**, **104**. The switches **102**, **104** are connected between an AC power source **106** and a lighting load **108**. The three-way switches **102**, **104** each include “movable” (or common) contacts, which are electrically connected to the AC power source **106** and the lighting load **108**, respectively. The three-way

switches **102**, **104** also each include two fixed contacts. When the movable contacts are making contact with the upper fixed contacts, the three-way switches **102**, **104** are in position A in FIG. 1A. When the movable contacts are making contact with the lower fixed contact, the three-way switches **102**, **104** are in position B. When the three-way switches **102**, **104** are both in position A (or both in position B), the circuit of system **100** is complete and the lighting load **108** is energized. When switch **102** is in position A and switch **104** is in position B (or vice versa), the circuit is not complete and the lighting load **108** is not energized.

Three-way dimmer switches that replace three-way switches are known in the art. An example of a three-way dimmer switch system **150**, including one prior art three-way dimmer switch **152** and one three-way switch **104** is shown in FIG. 1B. The three-way dimmer switch **152** includes a dimmer circuit **152A** and a three-way switch **152B**. A typical, AC phase-control dimmer circuit **152A** regulates the amount of energy supplied to the lighting load **108** by conducting for some portion of each half-cycle of the AC waveform, and not conducting for the remainder of the half-cycle. Because the dimmer circuit **152A** is in series with the lighting load **108**, the longer the dimmer circuit conducts, the more energy will be delivered to the lighting load **108**. Where the lighting load **108** is a lamp, the more energy that is delivered to the lighting load **108**, the greater the light intensity level of the lamp. In a typical dimming operation, a user may adjust a control to set the light intensity level of the lamp to a desired light intensity level. The portion of each half-cycle for which the dimmer conducts is based on the selected light intensity level. The user is able to dim and toggle the lighting load **108** from the three-way dimmer switch **152** and is only able to toggle the lighting load from the three-way switch **104**. Since two dimmer circuits cannot be wired in series, the three-way dimmer switch system **150** can only include one three-way dimmer switch **152**, which can be located on either the line side or the load side of the system.

A four-way switch system is required when there are more than two switch locations from which to control the load. For example, a four-way system requires two three-way switches and one four-way switch, wired in well known fashion, so as to render each switch fully operable to independently control the load irrespective of the status of any other switches in the system. In the four-way system, the four-way switch is required to be wired between the two three-way switches in order for all switches to operate independently, i.e., one three-way switch must be wired at the AC source side of the system, the other three-way switch must be wired at the load side of the system, and the four-way switch must be electrically situated between the two three-way switches.

FIG. 1C shows a prior art four-way switching system **180**. The system **180** includes two three-way switches **102**, **104** and a four-way switch **185**. The four-way switch **185** has two states. In the first state, node A1 is connected to node A2 and node B1 is connected to node B2. When the four-way switch **185** is toggled, the switch changes to the second state in which the paths are now crossed (i.e., node A1 is connected to node B2 and node B1 is connected to node A2). Note that a four-way switch can function as a three-way switch if one terminal is simply not connected.

FIG. 1D shows another prior art switching system **190** containing a plurality of four-way switches **185**. As shown, any number of four-way switches can be included between the three-way switches **102**, **104** to enable multiple location control of the lighting load **108**.

Multiple location dimming systems employing a smart dimmer and one or more specially-designed remote (or “accessory”) dimmers have been developed. The remote dimmers permit the intensity level of the lighting load to be adjusted from multiple locations. A smart dimmer is one that includes a microcontroller or other processing means for providing an advanced set of control features and feedback options to the end user. For example, the advanced features of a smart dimmer may include a protected or locked lighting preset, fading, and double-tap to full intensity. The microcontroller controls the operation of the semiconductor switch to thus control the intensity of the lighting load.

To power the microcontroller, the smart dimmers include power supplies, which draw a small amount of current through the lighting load when the semiconductor switch is non-conductive each half-cycle. The power supply typically uses this small amount of current to charge a storage capacitor and develop a direct-current (DC) voltage to power the microcontroller. An example of a multiple location lighting control system, including a wall-mountable smart dimmer switch and wall-mountable remote switches for wiring at all locations of a multiple location dimming system, is disclosed in commonly assigned U.S. Pat. No. 5,248,919, issued on Sep. 28, 1993, entitled LIGHTING CONTROL DEVICE, which is herein incorporated by reference in its entirety.

Referring again to the system 150 of FIG. 1B, since no load current flows through the dimmer circuit 152A of the three-way dimmer switch 152 when the circuit between the AC power source 106 and the lighting load 108 is broken by either three-way switch 152B or 104, the dimmer switch 152 is not able to include a power supply and a microcontroller. Thus, the dimmer switch 152 is not able to provide the advanced set of features of a smart dimmer to the end user.

FIG. 2 shows an example multiple location lighting control system 200 including one wall-mountable smart dimmer 202 and one wall-mountable remote dimmer 204. The dimmer 202 has a hot (H) terminal for receipt of an AC source voltage provided by an AC power source 206, and a dimmed-hot (DH) terminal for providing a dimmed-hot (or phase-controlled) voltage to a lighting load 208. The remote dimmer 204 is connected in series with the DH terminal of the dimmer 202 and the lighting load 208, and passes the dimmed-hot voltage through to the lighting load 208.

The dimmer 202 and the remote dimmer 204 both have actuators to allow for raising, lowering, and toggling on/off the light intensity level of the lighting load 208. The dimmer 202 is responsive to actuation of any of these actuators to alter the intensity level or to power the lighting load 208 on/off accordingly. In particular, an actuation of an actuator at the remote dimmer 204 causes an AC control signal, or partially rectified AC control signal, to be communicated from that remote dimmer 204 to the dimmer 202 over the wiring between the accessory dimmer (AD) terminal (i.e., accessory terminal) of the remote dimmer 204 and the AD terminal of the dimmer 202. The dimmer 202 is responsive to receipt of the control signal to alter the dimming level or toggle the load 208 on/off. Thus, the load can be fully controlled from the remote dimmer 204.

The user interface of the dimmer 202 of the multiple location lighting control system 200 is shown in FIG. 3. As shown, the dimmer 202 may include a faceplate 310, a bezel 312, an intensity selection actuator 314 for selecting a desired level of light intensity of a lighting load 208 controlled by the dimmer 202, and a control switch actuator 316. An actuation of the upper portion 314A of the actuator 314 increases or raises the light intensity of the lighting load 208,

while an actuation of the lower portion 314B of the actuator 314 decreases or lowers the light intensity.

The dimmer 202 may also include a visual display in the form of a plurality of light sources 318, such as light-emitting diodes (LEDs). The light sources 318 may be arranged in an array (such as a linear array as shown), and are illuminated to represent a range of light intensity levels of the lighting load 208 being controlled. The intensity levels of the lighting load 208 may range from a minimum intensity level, which may be the lowest visible intensity, but which may be “full off”, or 0%, to a maximum intensity level, which is typically “full on”, or substantially 100%. Light intensity level is typically expressed as a percent of full intensity. Thus, when the lighting load 208 is on, light intensity level may range from 1% to substantially 100%.

FIG. 4 is a simplified block diagram of the dimmer 202 and the remote dimmer 204 of the multiple location lighting control system 200. The dimmer 202 includes a bidirectional semiconductor switch 420, e.g., a triac or two field-effect transistors (FETs) in anti-series connection, coupled between the hot terminal H and the dimmed-hot terminal DH, to control the current through, and thus the light intensity of, the lighting load 208. The semiconductor switch 420 has a control input (or gate), which is connected to a gate drive circuit 424. The input to the gate renders the semiconductor switch 420 conductive or non-conductive, which in turn controls the power supplied to the lighting load 208. The gate drive circuit 424 provides control inputs to the semiconductor switch 420 in response to command signals from a microcontroller 426.

The microcontroller 426 receives inputs from a zero-crossing detector 430 and a signal detector 432 and controls the semiconductor switch 420 accordingly. The microcontroller 426 also generates command signals to a plurality of LEDs 418 for providing feedback to the user of the dimmer 202. A power supply 428 generates a DC output voltage V_{ec} to power the microcontroller 426. The power supply is coupled between the hot terminal H and the dimmed hot terminal DH.

The zero-crossing detector 430 determines the zero-crossings of the input AC supply voltage from the AC power supply 206. A zero-crossing is defined as the time at which the AC supply voltage transitions from positive to negative polarity (i.e., a negative-going zero-crossing), or from negative to positive polarity (i.e., a positive-going zero-crossing), at the beginning of each half-cycle. The zero-crossing information is provided as an input to microcontroller 426. The microcontroller 426 provides the gate control signals to operate the semiconductor switch 420 to provide voltage from the AC power source 206 to the lighting load 208 at predetermined times relative to the zero-crossing points of the AC waveform.

Generally, two techniques are used for controlling the power supplied to the lighting load 208: forward phase control dimming and reverse phase control dimming. In forward phase control dimming, the semiconductor switch 420 is turned on at some point within each AC line voltage half-cycle and remains on until the next voltage zero-crossing. Forward phase control dimming is often used to control energy to a resistive or inductive load, which may include, for example, a magnetic low-voltage transformer or an incandescent lamp. In reverse phase control dimming, the semiconductor switch 420 is turned on at the zero-crossing of the AC line voltage and turned off at some point within each half-cycle of the AC line voltage. Reverse phase control is often used to control energy to a capacitive load, which may include, for example, an electronic low-voltage

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transformer. Since the semiconductor switch **420** must be conductive at the beginning of the half-cycle, and be able to be turned off with in the half-cycle, reverse phase control dimming requires that the dimmer have two FETs in anti-serial connection, or the like.

The signal detector **432** has an input **440** for receiving switch closure signals from momentary switches T, R, and L. Switch T corresponds to a toggle switch controlled by the switch actuator **316**, and switches R and L correspond to the raise and lower switches controlled by the upper portion **314A** and the lower portion **314B**, respectively, of the intensity selection actuator **314**.

Closure of switch T connects the input of the signal detector **432** to the DH terminal of the dimmer **202**, and allows both positive and negative half-cycles of the AC current to flow through the signal detector. Closure of switches R and L also connects the input of the signal detector **432** to the DH terminal. However, when switch R is closed, current only flows through the signal detector **432** during the positive half-cycles of the AC power source **406** because of a diode **434**. In similar manner, when switch L is closed, current only flows through the signal detector **432** during the negative half-cycles because of a diode **436**. The signal detector **432** detects when the switches T, R, and L are closed, and provides two separate output signals representative of the state of the switches as inputs to the microcontroller **426**. A signal on the first output of the signal detector **432** indicates a closure of switch R and a signal on the second output indicates a closure of switch L. Simultaneous signals on both outputs represents a closure of switch T. The microprocessor controller **426** determines the duration of closure in response to inputs from the signal detector **432**.

The remote dimmer **204** provides a means for controlling the dimmer **202** from a remote location in a separate wall box. The remote dimmer **204** includes a further set of momentary switches T', R', and L' and diodes **434'** and **436'**. The wire connection is made between the AD terminal of the remote dimmer **204** and the AD terminal of the dimmer **202** to allow for the communication of actuator presses at the remote switch. The AD terminal is connected to the input **440** of the signal detector **432**. The action of switches T', R', and L' in the remote dimmer **204** corresponds to the action of switches T, R, and L in the dimmer **202**.

Since the remote dimmer **204** does not have LEDs, no feedback can be provided to a user at the remote dimmer **204**. Therefore there is a need for multiple location dimming system in which the remote devices include visual displays for providing feedback to a user.

SUMMARY

A load control system may be provided for controlling an amount of power delivered to an electrical load from an AC power source. The load control system may include a load control device and a remote device. The load control device may be adapted to be coupled between the AC power source and the electrical load to control the amount of power delivered to the electrical load. The load control device may include a first terminal and a second terminal, a first switching circuit coupled to the second terminal, and a second switching circuit coupled between the first and second terminals. The remote device may be adapted to be coupled to the first terminal and the second terminal of the load control device. The remote device may include a power supply configured to conduct a charging current from the AC power source through the first switching circuit of the load control device.

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The load control device may be configured to control the second terminal in an active pull-up state by rendering the first switching circuit conductive and the second switching circuit non-conductive. The load control device may be further configured to control the second terminal into an active pull-down state by rendering the second switching circuit conductive and the first switching circuit non-conductive. The load control device may be configured to render the first switching circuit conductive to allow the power supply of the remote device to charge through the second terminal during a first time period of a half cycle of the AC power source. Further, the load control device and the remote device may be operable to communicate with each other by controlling the second terminal into the active pull-up state and the active pull-down state during a second time period of the half-cycle of the AC power source.

A load control device may be used to controlling an amount of power delivered to an electrical load from an AC power source. The load control device may be adapted to be coupled to a remote device. The load control device may include a semiconductor switch configured to conduct a load current from the AC power source to the electrical load, a first terminal and a second terminal that are adapted to be coupled to the remote device, a first switching circuit, a second switching circuit, and a control circuit. The first switching circuit may be coupled to the second terminal, and may be configured to conduct a charging current from the AC power source to a power supply of the remote device. The second switching circuit may be coupled between the first terminal and the second terminal. The first terminal may be configured to be connected to the AC power source or the electrical load, and the second terminal may comprises (e.g., may be) an accessory terminal. The accessory terminal may be configured to be coupled to the remote device via accessory wiring.

The control circuit may be configured to control the semiconductor switch to control the power delivered to the electrical load. The control circuit may also be configured to render the first switching circuit conductive to conduct the charging current from the AC power source to the power supply of the remote device during a first time period of a half-cycle of the AC power source, and further configured to render the first and second switching circuits conductive and non-conductive to communicate with the remote device via the second terminal during a second time period of the half-cycle of the AC power source.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a diagram of an example of a prior art three-way switch system, which includes two three-way switches.

FIG. 1B is a diagram of an example of a prior art three-way dimmer switch system including one prior art three-way dimmer switch and one three-way switch.

FIG. 1C is a diagram of an example of a prior art four-way switching system.

FIG. 1D is a diagram of an example of a prior art extended four-way switching system.

FIG. 2 is a diagram of an example of a prior art multiple location lighting control system having a dimmer switch and a remote switch.

FIG. 3 is a front view of an example of a user interface of the dimmer switch of the multiple location lighting control system of FIG. 2.

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FIG. 4 is a diagram of an example of the dimmer switch and the remote switch of the multiple location lighting control system of FIG. 2.

FIG. 5A is a block diagram of an example of a multiple location load control system.

FIG. 5B is a block diagram of an example of a multiple location load control system.

FIG. 6 is a perspective view of an example of a user interface of a load control device.

FIG. 7 is a block diagram of an example main load control device of a multiple location system.

FIG. 8A is a block diagram of an example remote load control device of a multiple location system.

FIG. 8B is a block diagram of another example remote load control device of a multiple location system.

FIG. 9 is an example schematic diagram of the multi-location circuit and control circuit of the main load control device and the multi-location circuit and control circuit of the remote load control device of the system of FIG. 5A or FIG. 5B.

FIG. 10 is a timing diagram of an example of a complete line cycle of an AC voltage waveform provided by an AC power source.

FIG. 11 is a diagram of an example of a payload format for communication between the main load control device and the remote load control device of the system of FIG. 5A or FIG. 5B.

FIG. 12 is a flowchart of an example of a user interface procedure executed by the control circuit of the main load control device of FIG. 7.

FIG. 13 is a flowchart of an example of an Idle routine of the user interface procedure of FIG. 12.

FIG. 14 is a flowchart of an example of an ActiveHold routine of the user interface procedure of FIG. 12.

FIG. 15 is a flowchart of an example of a Release routine of the user interface procedure of FIG. 12.

FIG. 16 is a flowchart of an example of a RX buffer procedure executed by the control circuit of the main load control device of FIG. 7.

FIG. 17 is a flowchart of an example of a multi-location control procedure executed by the control circuit of the main load control device of FIG. 7.

FIG. 18 is a flowchart of an example of a multi-location control procedure executed by the control circuit of the remote load control device of FIG. 8A or FIG. 8B.

DETAILED DESCRIPTION

FIG. 5A is a block diagram of an example of a multiple location load control system, e.g., a multiple location dimming system 500. The multiple location dimming system 500 may comprise a main load control device, e.g., a main dimmer 502, and one or more remote load control devices, e.g., two remote dimmers 504 (i.e., accessory dimmers). The main dimmer 502 and remote dimmers 504 may be coupled in series electrical connection between an AC power source 506 and a lighting load 508, for example, via a traveler wiring 511. The traveler wiring 511 may couple the AC power source 506 to the lighting load 508 via the main dimmer 502 and one or more remote dimmers 504, for example, to provide power to the lighting load 508. Neutral wiring 512 may couple the lighting load 508 back to the AC power source 506, for example, to provide a return path for any remaining power provided by the AC power source 506 and not dissipated by the lighting load 508.

The main dimmer 502 may be wired to the line side of the system 500 (e.g., as shown) or the load side of the system

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500. Although the description herein is primarily with reference to the main dimmer 502 wired to the line side of the system 500, one or more embodiments may comprise the main dimmer 502 wired to the load side of the system 500 (e.g., and one or more remote dimmers 504 wired to the line side, accordingly). Further, any number of (e.g., more than two) remote dimmers 504 may be provided in the multiple location dimming system 500.

The main dimmer 502 may comprise a first main terminal and a second main terminal. For example, the main dimmer 502 may comprise a hot terminal H (i.e., a line-side load terminal) adapted to be coupled to the line-side of the system 500 and a dimmed-hot terminal DH (i.e., a load-side terminal) adapted to be coupled to the load-side of the system 500. The main dimmer 502 may comprise a load control circuit coupled between the hot and dimmed-hot terminals for controlling the amount of power delivered to the lighting load 508 (e.g., as described with reference to FIG. 7). The remote dimmers 504 may comprise a first main terminal and a second main terminal. For example, the remote dimmers 504 may comprise two hot terminals H1, H2, which may conduct the load current from the AC power source 506 to the lighting load 508. The main dimmer 502 and the remote dimmers 504 may each comprise an internal air-gap switch (e.g., air-gap switches 722, 822 shown in FIGS. 7 and 8A) for disconnecting the lighting load 508 from the AC power source 506. The main dimmer 502 and the remote dimmers 504 may each comprise an accessory dimmer (AD) terminal AD (i.e., accessory terminal) coupled together via a single accessory dimmer (AD) line 509 (i.e., an accessory wiring). The main dimmer 502 and the remote dimmers 504 may be operable to communicate, i.e., transmit and receive digital messages, via the AD line 509. The main dimmer 502 and the remote dimmers 504 may not include connections to the neutral side of the AC power source 506.

The main dimmer 502 and the remote dimmer 504 may include actuators and visual displays, such that lighting load 508 may be controlled from and feedback of the lighting load may be provided at each of the main dimmer 502 and the remote dimmers 504. In order to provide the visual displays at the remote dimmers 504, the remote dimmers 504 may include a control circuit (e.g., which may comprise a microprocessor) and a power supply for powering the microprocessor. The main dimmer 502 may provide an AD supply voltage VAD (e.g., approximately 80-170 VDC) on the AD line 509 to enable the power supplies of the remote dimmers 504 to charge during a first portion (i.e., a charging time T_{CHRG}) of a half-cycle of the AC power source 506. During a second portion (i.e., a communication time T_{COMM}) of the half-cycle, the main dimmer 502 and the remote dimmers 504 are operable to transmit and receive the digital messages via the AD line 509.

FIG. 5B is a block diagram of an example of a multiple location load control system, e.g., a multiple location dimming system 510. The multiple location dimming system 510 may comprise a main load control device, e.g., a main dimmer 502, and one or more remote load control devices, e.g., two remote dimmers 514 (i.e., accessory dimmers). The remote dimmers 514 may be substantially similar to the remote dimmers 504, except the remote dimmers 514 may comprise a single main terminal (e.g., a single hot terminal H') as opposed to the first and second hot terminals H1 and H2 and may not comprise an air-gap switch (e.g., the air-gap switch 822 shown in FIG. 8A). One or more of the embodiments described herein with reference to the multiple location dimming system 500 and/or the remote dimmers 504

may be applicable to the multiple location dimming system **510** and/or the remote dimmers **514**.

The main dimmer **502** may be coupled in series electrical connection between the AC power source **506** and the lighting load **508**, for example, via traveler wiring **511**. The traveler wiring **511** may couple the AC power source **506** to the lighting load **508** via the main dimmer **502**, for example, to provide power to the lighting load **508**. The one or more remote dimmers **514** may be coupled to the traveler wiring **511** via the hot terminal H'. Neutral wiring **512** may couple the lighting load **508** back to the AC power source **506**, for example, to provide a return path for any remaining power provided by the AC power source **506** and not dissipated by the lighting load **508**. The main dimmer **502** may be wired to the line side of the system **510** (e.g., as shown) or the load side of the system **510**. Although the description herein is primarily with reference to the main dimmer **502** wired to the line side of the system **510**, one or more embodiments may comprise the main dimmer **502** wired to the load side of the system **510** (e.g., and one or more remote dimmers **514** wired to the line side, accordingly). Further, any number of (e.g., more than two) remote dimmers **514** may be provided in the multiple location dimming system **510**.

The hot terminal H' of the remote dimmers **514** may be connected to the dimmed hot terminal DH of the main dimmer **502** (e.g., as shown) and to the lighting load **508** via the traveler wiring **511**, for example, if the main dimmer **502** is wired to the line side of the system **510**. If the main dimmer **502** is wired to the load side of the system **510**, then the hot terminal H' of the remote dimmers **514** may be connected to the hot terminal H of the main dimmer **502** and to the AC power source **506** via the traveler wiring **511**. The main dimmer **502** and the remote dimmers **514** may each comprise accessory dimmer terminals AD (i.e., accessory terminals) coupled together via a single accessory dimmer (AD) line **509** (i.e., an accessory wiring). The main dimmer **502** and the remote dimmers **514** may be operable to communicate, i.e., transmit and receive digital messages, via the AD line **509**. The main dimmer **502** and the remote dimmers **514** may not include connections to the neutral side of the AC power source **506**.

The main dimmer **502** and the remote dimmer **514** may include actuators and visual displays, such that lighting load **508** may be controlled from and feedback of the lighting load may be provided at each of the main dimmer **502** and the remote dimmers **514**. In order to provide the visual displays at the remote dimmers **514**, the remote dimmers **514** may include a control circuit (e.g., which may comprise a microprocessor) and a power supply for powering the microprocessor. The main dimmer **502** may provide an AD supply voltage VAD (e.g., approximately 80-170 VDC) on the AD line **509** to enable the power supplies of the remote dimmers **514** to charge during a first portion (i.e., a charging time T_{CHRG}) of a half-cycle of the AC power source **506**. During a second portion (i.e., a communication time T_{COMM}) of the half-cycle, the main dimmer **502** and the remote dimmers **514** are operable to transmit and receive the digital messages via the AD line **509**.

FIG. 6 is a diagram of an example user interface **600** of a load control device, which may be provided on, for example, the main dimmer **502** and/or the remote dimmers **504**, **514** of the multiple location dimming system **500** shown in FIG. 5A and/or the multiple location dimming system **510** shown in FIG. 5B. The user interface **600** may include a thin touch sensitive actuator **610** comprising an actuation member **612** having first and second portions **612A**, **612B**. The actuation member **612** may extend through

a bezel **614** to contact a touch sensitive device (not shown) located inside the main dimmer **502** and the remote dimmers **504**, **514**. The main dimmer **502** may be operable to control the intensity of a connected lighting load **508** in response to actuations of the actuation member **612** of either the main dimmer **502** or the remote dimmers **504**, **514**.

The user interface **600** may comprise a faceplate **616**, which may include a non-standard opening **618** and may mount to an adapter **620**. The bezel **614** may be housed behind the faceplate **616** and extend through the opening **618**. The adapter **620** may connect to a yoke (not shown), which may be adapted to mount the main dimmer **502** and the remote dimmers **504**, **514** to standard electrical wall-boxes. An air-gap actuator **622** may allow for actuation of an internal air-gap switch (e.g., an internal air-gap switch **722** as shown in FIG. 7) by pulling the air-gap actuator **622** down.

The bezel **614** may comprise a break **624**, which may separate the lower portion **612A** and the upper portion **612B** of the actuation member **612**. Upon actuation of the lower portion **612B** of the actuation member **612**, the main dimmer **502** may cause the connected lighting load **508** to toggle from on to off, and vice versa. Actuation of the upper portion **612A** of the actuation member **612**, i.e., above the break **624**, may cause the intensity of the lighting load **508** to change to a level dependent upon the position of the actuation along the length of the actuation member **612**.

A plurality of visual indicators, e.g., a plurality of light-emitting diodes (LEDs), may be arranged in a linear array behind the actuation member **612**. The actuation member **612** may be substantially transparent, such that the LEDs are operable to illuminate portions of the actuation member. Two different color LEDs may be located behind the lower portion **612B**, such that the lower portion is illuminated, for example, with white light when the lighting load **508** is on and with orange light with the lighting load is off. The LEDs behind the upper portion **612A** may be, for example, white and may be illuminated as a bar graph to display the intensity of the lighting load **508** when the lighting load is on.

The touch sensitive actuator **610** of the user interface **600** may be described in greater detail in commonly-assigned U.S. Pat. No. 7,791,595, issued Sep. 7, 2010, entitled TOUCH SCREEN ASSEMBLY FOR A LIGHTING CONTROL, the entire disclosure of which is hereby incorporated by reference.

FIG. 7 is a block diagram of an example main load control device of a multiple location load control system, e.g., the main dimmer **502**. The main dimmer **502** may comprise a bidirectional semiconductor switch **710**, a gate drive circuit **712**, a control circuit **714**, a zero-crossing detector **716**, a memory **718**, an audible sound generator **720**, an air-gap switch **722**, an inductor **724**, a communication circuit **725**, the user interface **600**, and/or a multi-location circuit **732**.

The main dimmer **502** may employ the bidirectional semiconductor switch **710** (e.g., a triac) coupled between the hot terminal H and the dimmed hot terminal DH, to control the current through, and thus the intensity of, the lighting load **508**. The semiconductor switch **710** may be implemented as any suitable bidirectional semiconductor switch, such as, for example, a FET in a full-wave rectifier bridge, two FETs in anti-series connection, or one or more insulated-gate bipolar junction transistors (IGBTs). The semiconductor switch **710** may comprise a control input (e.g., gate), which is connected to the gate drive circuit **712**. The input to the gate may render the semiconductor switch **710**

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selectively conductive or non-conductive, which in turn may control the power supplied to the lighting load **508**.

The control circuit **714** may be operable to control the semiconductor switch **710** by providing a control signal to the gate drive circuit **712** using the forward phase control dimming technique and/or the reverse phase control dimming technique. For example, the control circuit **714** may comprise a microcontroller, a microprocessor, a programmable logic device (PLD), a field programmable grid array (FPGA), an application specific integrated circuit (ASIC), or any suitable processing device, controller, or control circuit. The control circuit **714** may be coupled to a zero-crossing detect circuit **716**, which may determine the zero-crossing points of the AC line voltage from the AC power supply **506**. The control circuit **714** may generate the gate control signals to operate the semiconductor switch **710** to thus provide voltage from the AC power supply **506** to the lighting load **508** at predetermined times relative to the zero-crossing points of the AC line voltage.

The user interface **600** may be coupled to the control circuit **714**, such that the control circuit **714** is operable to receive inputs from the touch sensitive actuator **610** and to control the LEDs to provide feedback of the amount of power presently being delivered to the lighting load **508**. An example of the electrical circuitry of the user interface **600** may be described in greater detail in co-pending, commonly-assigned U.S. Pat. No. 7,855,543, issued Dec. 21, 2010, entitled FORCE INVARIANT TOUCH SENSITIVE ACTUATOR, the entire disclosure of which is hereby incorporated by reference.

The main dimmer **502** may further comprise an audible sound generator **718** coupled to the control circuit **714**. The control circuit **714** may be operable to cause the audible sound generator **718** to produce an audible sound in response to an actuation of the touch sensitive actuator **610**. A memory **718** may be coupled to the control circuit **714** and may be operable to store control information of the main dimmer **502**.

The air-gap switch **722** may be coupled in series between the hot terminal H and the semiconductor switch **710**. The air-gap switch **722** may have a normally-closed state in which the semiconductor switch **710** is coupled in series electrical connection between the AC power source **506** and the lighting load **508**. When the air-gap switch **722** is actuated (i.e., in an open state), the air-gap switch may provide an actual air-gap break between the AC power source **506** and the lighting load **508**. The air-gap switch **722** may allow a user to service the lighting load **508** without the risk of electrical shock. The main dimmer **502** may comprise the inductor **724** (i.e., a choke) for providing electromagnetic interference (EMI) filtering.

The main dimmer **502** may comprise a power supply **730** for generating a DC supply voltage V_{CC} (e.g., approximately 3.3 volts) for powering the control circuit **714** and other low voltage circuitry of the main dimmer **502**. The power supply **730** may draw (e.g., only draw) current at the beginning of a half-cycle (e.g., each half-cycle) while the bidirectional semiconductor switch **710** is non-conductive, for example, if the forward phase control dimming technique is used. The power supply **730** may draw (e.g., only draw) current at the end (i.e., trailing edge) of a half-cycle (e.g., each half-cycle) while the bidirectional semiconductor switch **710** is non-conductive, for example, if the reverse phase control dimming technique is used. The power supply **730** may stop drawing current when the bidirectional semiconductor switch **710** is rendered conductive.

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The multi-location circuit **732** may be coupled between the hot terminal H and/or the dimmed hot terminal DH and an accessory dimmer terminal AD (which may be adapted to be coupled to the AD line **509**). The multi-location circuit **732** may provide a supply voltage to the remote dimmer **504**, **514** via the AD line **509** and/or allow for communication of a digital message between the main dimmer **502** and the remote dimmers **504**, **514** via the AD line **509**. The control circuit **714** may provide a control signal to the multi-location circuit. If the main dimmer **502** is located on the line side of the system **500/510**, then the control circuit **714** may control the multi-location circuit **732** to allow the remote dimmers **504**, **514** to charge their internal power supplies and transmit and receive digital messages during the positive half-cycles. If the main dimmer **502** is located on the load side of the system **500/510**, then the control circuit **714** may control the multi-location circuit **732** to allow the remote dimmers **504**, **514** to charge their internal power supplies and transmit and receive digital messages during the negative half-cycles.

The main dimmer **502** may comprise another communication circuit **725** (e.g., in addition to the multi-location circuit **732**) for transmitting or receiving digital messages via a communications link, for example, a wired serial control link, a power-line carrier (PLC) communication link, or a wireless communication link, such as an infrared (IR) or a radio-frequency (RF) communication link. An example of a load control device able to transmit and receive digital messages on an RF communication link is described in commonly assigned U.S. Pat. No. 5,905,442, issued May 18, 1999, entitled METHOD AND APPARATUS FOR CONTROLLING AND DETERMINING THE STATUS OF ELECTRICAL DEVICES FROM REMOTE LOCATIONS, the entire disclosure of which is hereby incorporated by reference.

FIG. 8A is a diagram of an example remote load control device of a multiple location load control system, e.g., the remote dimmer **504** of the load control system **500** shown in FIG. 5A. The remote dimmer **504** may comprise one or more of the same functional blocks as the main dimmer **502**. The remote dimmer **504** may comprise a control circuit **814**, a zero-crossing detector **816**, an air-gap switch **822**, a multi-location circuit **832**, a power supply **830**, the user interface **600**, the memory **718**, the audible sound generator **720**, and/or the communication circuit **725**.

The control circuit **814** may comprise a microcontroller, a microprocessor, a programmable logic device (PLD), a field programmable grid array (FPGA), an application specific integrated circuit (ASIC), or any suitable processing device, controller, or control circuit. The control circuit **814** may be coupled to a zero-crossing detect circuit **816**, which may determine the zero-crossing points of the AC line voltage from the AC power supply **506**. The user interface **600** may be coupled to the control circuit **814**, such that the control circuit **814** is operable to receive inputs from the touch sensitive actuator **610** and to control the LEDs to provide feedback of the amount of power presently being delivered to the lighting load **508**.

The remote dimmer **504** may comprise first and second hot terminals H1, H2 that may be coupled in series with the bidirectional semiconductor switch **710** of the main dimmer **502**, and may be adapted to conduct the load current from the AC power source **506** to the lighting load **508**. The remote dimmer **504** may also comprise an accessory dimmer terminal AD that is adapted to be coupled to the accessory dimmer terminal AD of the main dimmer **502** via the AD line **509**.

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The power supply **830** may be coupled between the multi-location circuit **832** and the first and second hot terminals H1, H2 to draw power from the main dimmer **502**, via the multi-location circuit **832**, during the charging time period T_{CHRG} of a half-cycle. The power supply **830** may generate a DC output voltage V_{DD} (e.g., approximately 3.3 volts) for powering the control circuit **814** and other low voltage circuitry of the remote dimmer **504**. The power supply **830** may comprise a capacitor **940**, for example, as shown in FIG. 9.

The zero-crossing detector **816** may be coupled between the accessory dimmer terminal AD and the first and second hot terminals H1, H2. The zero-crossing detector **816** may detect a zero-crossing and/or may couple the AD supply voltage VAD across the zero-crossing detector **816**. The control circuit **814** may begin timing at a zero-crossing (e.g., each zero-crossing) and may be operable to transmit and receive digital messages via the multi-location circuit **832**, for example, after the charging time period T_{CHRG} expires. The multi-location circuit **832** may be coupled between the AD line **509** and the power supply **830**. The multi-location circuit **832** and power supply **830** of the remote dimmer **504** may be coupled in parallel with the multi-location circuit **732** of the main dimmer **502** forming a communication path during the communication time period T_{COMM} in the positive and/or negative half-cycles, for example, depending on which side of the system **500/510** to which the main dimmer **502** is coupled. Accordingly, the communication path between the main dimmer **502** and the remote dimmer **504** may not pass through the AC power source **506** or the lighting load **508**.

The opening of the air-gap switch **822** of the remote dimmer **504** may provide a true air-gap disconnect between the AC power source **506** and the lighting load **508**. The zero-crossing detector **816**, the power supply **830**, and the multi-location circuit **832** of the remote dimmer **504** may include diodes coupled to the accessory dimmer terminal AD, such that the accessory dimmer terminal AD of the remote dimmer **504** may be operable (e.g., only operable) to conduct current into the remote dimmer **504**. The path for leakage current through the system **500** may be through the dimmed hot terminal DH and out of the accessory dimmer terminal AD of the main dimmer **502**. The orientation of the first and second hot terminals H1 and H2 of the remote dimmer **504** with respect to the main dimmer **502** may be reversed, for example, such that the second hot terminal H2 of the remote dimmer **504** may be coupled to the dimmed hot terminal DH of the main dimmer **502** and the first hot terminal H1 of the remote dimmer **504** may be coupled to the lighting load **508**. This may be performed to the path for leakage current to the lighting load **508** through the accessory dimmer terminal AD of the remote dimmer **504**. The components chosen for these circuits may be such that the magnitude of the leakage current through the main dimmer **502** is limited to an appropriate level to meet the UL standard for leakage current when the air-gap switch **722** is opened.

When any of the main dimmer **502** and the remote dimmers **504** are wired directly to the AC power source **506** and the lighting load **508**, the respective air-gap switches **722**, **822** may be positioned towards the AC power source and the lighting load, such that opening the air-gap switches **722**, **822** may provide a true air-gap disconnect between the AC power source **506** and the lighting load **508**. However, if any of the main dimmer **502** and the remote dimmers **504** that are wired directly to the AC power source **506** and the lighting load **508** do not have their air-gap switches **722**, **822**

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positioned towards the AC power source **506** and the lighting load **508**, the leakage current through the main dimmer **502** and the remote dimmers **504** may be limited to meet the UL standard for leakage current when an air-gap switch is opened. The leakage current may be limited in this way when the air-gap switches **722**, **822** of any of the remote dimmers **504** that are wired in the middle of the system **500** are opened.

FIG. 8B is a diagram of another example remote load control device of a multiple location load control system, e.g., the remote dimmer **514** of the load control system **510** shown in FIG. 5B. The remote dimmer **514** may comprise one or more of the same functional blocks as the remote dimmer **504**. The remote dimmer **514** may comprise a control circuit **814**, a zero-crossing detector **816**, a multi-location circuit **832**, a power supply **830**, the user interface **600**, the memory **718**, the audible sound generator **720**, and/or the communication circuit **725**. One or more of the embodiments described herein with reference to the remote dimmers **504** (e.g., those associated with FIG. 9-18) may be applicable to the remote dimmers **514**.

The remote dimmer **514** may not comprise an air-gap switch **822**. As such, the remote dimmer **514** may comprise a single hot terminal H' as opposed to the first and second hot terminals H1 and H2. The single hot terminal H' of the remote dimmer **514** may be connected to the dimmed hot terminal DH of the main dimmer **502** and to the lighting load **508** (e.g., and the single H terminal of one or more additional remote dimmers **514**), for example, as illustrated in FIG. 5B. Alternatively, the single hot terminal H' may be connected to the hot terminal H of the main dimmer **502** and the AC power source **506**, for example, if the main dimmer **502** is wired to the line side. The single hot terminal H' of the remote dimmer **514** may be coupled to the bidirectional semiconductor switch **710** of the main dimmer **502**, for example, via the H or DH terminal. The remote dimmer **514** may not be adapted to conduct the load current from the AC power source **506** to the lighting load **508**, since for example, the DH terminal of the main dimmer **502** may be connected directly to the lighting load **508** (e.g., without traveling through the remote dimmer **514**). The accessory dimmer terminal AD of the remote dimmer **514** may be adapted to be coupled to the accessory dimmer terminal AD of the main dimmer **502** via the AD line **509**.

FIG. 9 is an example schematic diagram of a multi-location circuit and a control circuit, e.g., the multi-location circuit **732** and control circuit **714** of the main dimmer **502**, and/or the multi-location circuit **832** and control circuit **814** of the remote dimmer **504** and/or the remote dimmer **514**. Although described with respect to the remote dimmer **504**, the description of FIG. 9 may be applicable to the remote dimmer **514**, either entirely or in part. The main dimmer **502** may be connected to the remote dimmers **504** via the AD line **509** and the traveler line **511**. The traveler line **511** may be connected between the hot terminal or dimmed hot terminal of the main dimmer **502** and one of the hot terminals of the remote dimmer **504**, for example, depending on whether the main dimmer **502** is configured on the line side or the load side of the system **500**. The main dimmer **502** may communicate (i.e., transmit and receive digital messages) and deliver power to a remote dimmer **504** via the AD line **509**.

The multi-location circuit **732** of the main dimmer **502** may comprise an NPN bipolar junction transistor Q906, an NPN bipolar junction transistor Q908, an NPN bipolar junction transistor Q910, an NPN bipolar junction transistor Q912, a capacitor **918**, a diode D920, a resistor R922, a

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resistor R924, and/or a resistor R926. The control circuit 714 of the main dimmer 502 may comprise a universal asynchronous receiver/transmitter (UART) 928 and/or an NPN bipolar junction transistor Q916. The UART 928 may be an internal circuit of a microprocessor of the control circuit 714.

The collector of the transistor Q906 may be connected to the hot terminal of the main dimmer 502. The emitter of the transistor Q906 may be connected to the non-isolated circuit common and may be connected to the collector of the transistor Q908 through the capacitor 918. The collector of the transistor Q908 may be connected to the non-isolated circuit common through the capacitor 918 and the emitter of the transistor Q908 may be connected to the AD line 509 through the diode D920. The base of the transistor Q908 may be connected to the transmit node of the UART 928. The collector of the transistor Q910 may be connected to the AD line 509 and the emitter of the transistor Q910 may be connected to the non-isolated circuit common through the resistors R922, R924. The junction of the resistors R922, R924 may be coupled to the receive node (Rx) of the UART 928. The collector of the transistor Q912 may be connected to the AD line 509 and the emitter of the transistor Q912 may be connected to the non-isolated circuit common through the resistor R926.

The multi-location circuit 732 of the remote dimmer 504 may comprise an NPN bipolar junction transistor Q930, an NPN bipolar junction transistor Q932, an NPN bipolar junction transistor Q934, a diode D942, a resistor R944, a resistor R946, and/or a resistor R948. The control circuit 814 of the remote dimmer 504 may comprise a universal asynchronous receiver/transmitter (UART) 950 and an NPN bipolar junction transistor Q936. The remote dimmer 504 may comprise a capacitor 940, which may be coupled across the input of the power supply 830 of the remote dimmer 504. As such, the capacitor 940 may be coupled between the multi-location circuit 832 and a hot terminal H1/H2/H' to charge through the diode D942 from the main dimmer 502, via the multi-location circuit 832, during the charging time period T_{CHRG} of a half-cycle. The power supply 830 may conduct current from the main dimmer 502 and/or from the capacitor 940 to generate the DC supply voltage V_{DD} for powering the control circuit 1114 and other low voltage circuitry of the remote dimmer 504.

The collector of the transistor Q930 may be connected to the AD line 509 and the emitter of the transistor Q930 may be connected to the non-isolated circuit common through the resistor R944. The collector of the transistor Q932 may be connected to the AD line 509 and the emitter of the transistor Q932 may be connected to the non-isolated circuit common through the resistors R946, R948. The junction of the resistors R946, R948 may be coupled to the receive node (Rx) of the UART 950. The collector of the transistor Q934 may be connected to the capacitor 940 and the emitter of the transistor Q934 may be connected to the AD line 509. The base of the transistor Q934 may be connected to the transmit node of the UART 950.

The main dimmer 502 and/or the remote dimmer 504 may control the AD line 509 using tri-state logic. Tri-state logic may be referred to as tri-state communication, three-state logic, 3-state logic, and/or the like. The sender (e.g., the main dimmer 502 or the remote dimmer 504) may controlling the AD line 509 into one of three states, an active pull-up state, and active pull-down state, or a high impedance state. The main dimmer 502 and/or the remote dimmer 504 may control the AD line 509 using tri-state logic to, for

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example, charge a power supply (e.g., capacitor 940) of the remote dimmer 504 and/or communicate with one another.

The main dimmer 502 may charge the capacitor 940 of the remote dimmer 504 during a half-cycle of an AC voltage waveform (e.g., as shown in FIG. 10) using the AD line 509. The main dimmer 502 may actively pull up the AD line 509 to generate an AD supply voltage VAD on the AD line 509 during the changing time period T_{CHRG} . In the active pull up state, the potential between the AD line 509 and the traveler line 511 may vary between approximately 80 to 170 volts. To pull up the AD line 509 during the charging time period T_{CHRG} , the transistor Q906 and the transistor Q908 of the main dimmer 502 may be rendered conductive, while the transistor Q910 and the transistor Q912 of the main dimmer 502, and the transistor Q930, the transistor Q932, and the transistor Q934 of the remote dimmer 504 may not be rendered conductive. As such, a current from the AC power source 506 may be conducted through the transistor Q906 to charge the capacitor 918 of the main dimmer 502. In addition, a charging current may be conducted from the capacitor 918 through the transistor Q908, the diode D920, the AD line 509, and the diode D942 to charge the capacitor 940 of the remote dimmer 504. Therefore, the capacitor 940 of the remote dimmer 504 may be charged by the main dimmer 502 via the AD line 509 and the multi-location circuit 832 and the power supply 830 may generate the DC supply voltage V_{DD} . Residual current from the AC power source 506 provided after the capacitor 940 is fully charged may return via the traveler wire 511.

The main dimmer 502 and the remote dimmer 504 may communicate during a half-cycle of an AC voltage waveform (e.g., as shown in FIG. 10) using the AD line 509. For example, the main dimmer 502 may charge the capacitor 940 of the remote dimmer 504, and the main dimmer 502 and the remote dimmer 504 may communicate at least a portion of a digital message during a single half-cycle of an AC voltage waveform.

The main dimmer 502 and the remote dimmers 504 may communicate with one another by controlling the AD line 509. For example, the main dimmer 502 and/or the remote dimmer 504 may communicate by placing the AD line 509 in an active pull-up state and/or an active pull-down state. The receiver (e.g., the main dimmer 502 or the remote dimmer 504) may interpret a "1" bit when the AD line 509 is in the active pull-up state, a "0" bit when the AD line 509 is in the active pull-down state, and nothing when the AD line 509 is in the high impedance state. In the active pull up state, the potential between the AD line 509 and the traveler wire 511 may vary between approximately 80 to 170 volts. In the active pull down state, there may be no potential between the AD line 509 and the traveler wire 511. In the high impedance state, the potential between the AD line 509 and the traveler wire 511 may depend on the charge stored by the line capacitance of the electrical wiring between the main dimmer 502 and the remote dimmer 504, i.e., the AD line 509. The use of the active pull-up state and active-pull down state may allow for faster and/or more reliable communication, for example, because the active pull-up state and active-pull down state may be characterized by sharper edges between communications.

When the main dimmer 502 is transmitting a digital message to the remote dimmer 504, the transistor Q906 of the main dimmer 502 may be rendered conductive. To receive a digital message from the main dimmer 502, the remote dimmer 504 may render the transistor Q932 conductive (e.g., via the Rx_Enable line at the base of the transistor Q932). To place the AD line 509 in the active pull up state,

the main dimmer **502** may render the transistor **Q908** conductive and the transistor **Q912** non-conductive. As such, the AD line **509** is pulled up (i.e., pulled-up to approximately 80-170 volts) and the remote dimmer **504** (i.e., the UART **950** of the control circuit **814**) interprets a "1" bit being communicated. To place the AD line **509** in the active pull down state, the main dimmer **502** may render the transistor **Q908** non-conductive and the transistor **Q912** conductive. As such, the AD line **509** is pulled down and has substantially the same voltage potential as the traveler wire **511**. When the AD line **509** is pulled-down, the remote dimmer **504** interprets a "0" bit being communicated. For example, the transistor **Q912** may be rendered conductive (e.g., only rendered conductive) to transmit a "0" bit. Therefore, the main dimmer **502** may render a first switching circuit (e.g., transistor **Q908**) and a second switching circuit (e.g., transistor **Q912**) conductive and non-conductive on a complementary basis to transmit a digital message to the remote dimmer **504** via the AD line **509** during the communication period T_{COMM} of the half-cycle of the AC power source. For example, during communication, the main dimmer **502** may actively pull-up or pull-down the AD line **509** to communicate a "1" bit or a "0" bit, respectively, by rendering the transistors **Q908**, **Q912** conductive or non-conductive on a complementary basis.

When the remote dimmer **504** is transmitting a digital message to the main dimmer **502**, the transistor **Q934** of the remote dimmer **504** may be rendered conductive. To receive a digital message from the remote dimmer **504**, the main dimmer **502** may render the transistor **Q910** conductive (e.g., via the Rx_Enable line at the base of the transistor **Q910**). To place the AD line **509** in the active pull up state, the remote dimmer **504** may render the transistor **Q934** conductive and the transistor **Q930** non-conductive. As such, the AD line **509** is pulled up (i.e., pulled-up to approximately 80-170 volts) and the main dimmer **502** (i.e., the UART **928** of the control circuit **714**) interprets a "1" bit being communicated. To place the AD line **509** in the active pull down state, the remote dimmer **504** may render the transistor **Q934** non-conductive and the transistor **Q930** conductive. As such, the AD line **509** is pulled down and has substantially the same voltage potential as the traveler wire **511**. When the AD line **509** is pulled-down, the main dimmer **502** interprets a "0" bit being communicated. For example, the transistor **Q930** may be rendered conductive (e.g., only rendered conductive) to transmit a "0" bit. Therefore, the remote dimmer **504** may render a first switching circuit (e.g., transistor **Q934**) and a second switching circuit (e.g., transistor **Q930**) conductive and non-conductive on a complementary basis to transmit a digital message to the main dimmer **502** via the AD line **509** during the communication period T_{COMM} of the half-cycle of the AC power source. For example, during communication, the remote dimmer **504** may actively pull-up or pull-down the AD line **509** to communicate a "1" bit or a "0" bit, respectively, by rendering the transistors **Q930**, **Q934** conductive or non-conductive on a complementary basis.

The AD line **509** may be placed in the high impedance state. To place the AD line **509** in the high impedance state, the transistor **Q906** may be rendered conductive and the transistors **Q908**, **Q910**, and **Q912** of the main dimmer **502** and the transistors **Q930**, **Q932**, and **Q934** of the remote dimmer may be rendered non-conductive. As such, in the high impedance state, the potential between the AD line **509** and the traveler wire **511** may depend on the charge stored by the AD line **509**. The interpretation of the AD line **509** by the receiver (e.g., the main dimmer **502** or the remote

dimmer **504**) in the high impedance state is indeterminable. The multi-location circuits **732**, **832** dissipate less power in the high impedance state.

In the multi-location circuit **732**, the Pull down line **960** of the transistor **Q912** may be the inverted version of the Tx_control line **962** at the base of the transistor **Q908**. For example, an inverter circuit (not shown) may be located between the base of the transistor **Q912** (i.e., on the Pull down line **960**) and the base of the transistor **Q908** (i.e., on the Tx_control line **962**). The Pull down line **960** also may be coupled (not shown) to the control circuit **714** (e.g., to an open drain output of a microprocessor), so that the control circuit **714** may pull down the base of the transistor **Q912** to render the transistor **Q912** non-conductive during the high impedance state (i.e., to disable control of the transistor **Q912** in response to the Pull down line **960**). The transistor **Q912** may be rendered conductive (e.g., only rendered conductive) when the transistor **Q908** is rendered non-conductive during the communication time.

In the multi-location circuit **832**, the Pull down line **968** of the transistor **Q930** may be the inverted version of the Tx_control line **966** at the base of the transistor **Q934**. For example, an inverter circuit (not shown) may be located between the base of the transistor **Q930** (i.e., on the Pull down line **968**) and the base of the transistor **Q934** (i.e., on the Tx_control line **966**). The Pull down line **968** may be coupled (not shown) to the control circuit **814** (e.g., to an open drain output of a microprocessor), so that the control circuit **814** may pull down the base of the transistor **Q930** to render the transistor **Q930** non-conductive during the high impedance state (i.e., to disable control of the transistor **Q912** in response to the Pull down line **968**). The transistor **Q930** may be rendered conductive (e.g., only rendered conductive) when the transistor **Q934** is rendered non-conductive during the communication time.

FIG. **10** is a timing diagram of an example of a complete line cycle of an AC voltage waveform **1000** provided by an AC power source (e.g., the AC power source **506**). The timing diagram of FIG. **10** illustrates an example of the operation of a main dimmer **502** during each half-cycle of the AC voltage waveform **1000**. The main dimmer **502** may be operable to allow one or more remote dimmers **504** connected to the AD line **509** to charge their internal power supplies (i.e., capacitor **940**) during a charging time period T_{CHRG} . The charging time period T_{CHRG} may occur after a zero-crossing **1002** at the beginning of the positive half-cycle of the AC voltage waveform **1000**. The charging time period T_{CHRG} may be approximate 2 ms in duration. The AD line **509** may be pulled up by the main dimmer **502** during the charging time period T_{CHRG} to charge the capacitors **940** of the remote dimmers **504**, for example, as described with reference to FIG. **9**.

After the charging time period T_{CHRG} , a first buffer time T_{BUF1} may be used to ensure that the state of the AD line **509** during the charging time period T_{CHRG} is not misinterpreted as part of a digital message during the communication time period T_{COMM} .

After the buffer time T_{BUF1} , the main dimmer **502** and one or more of the remote dimmers **504** may be operable to transmit and receive digital messages via the AD line **509** during the communication time period T_{COMM} . The communication time period T_{COMM} may occur after the buffer time T_{BUF1} and during the positive half-cycle of the AC voltage waveform **1000**. The communication time period T_{COMM} may be approximate 3.75 ms. The communication time period T_{COMM} may be a dedicated time slot for communication between the main dimmer **502** and one or more

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remote dimmers **504**. The main dimmer **502** and/or a remote dimmer **504** may pull up and/or pull down the AD line **509** to transmit a digital message, for example, as described with reference to FIG. 9. As such, communication between the main dimmer **502** and one or more remote dimmers **504** may be performed during the communication time period T_{COMM} using the active pull-up state and/or the active-pull down state. After the communication time period T_{COMM} , the AD line **509** may be in a high impedance state (i.e., the high impedance state).

The remote dimmer **504** may monitor for the beginning of a charge pulse during a charge pulse window T_{CPW} right before the next zero-crossing **1006**. The charge pulse may occur during the charging time period T_{CHRG} each line cycle. The charge pulse window T_{CPW} may begin after a charge pulse window delay period T_{DELAY} , which may have a duration of approximately 14 ms measured from the zero-crossing **1002**. The charge pulse window T_{CPW} may begin at a time **1005** before the zero-crossing **1006** between the negative half-cycle of the AC voltage waveform **1000** and a subsequent cycle of the AC voltage waveform **1000**, for example, as shown in FIG. 10. During the charge pulse window T_{CPW} , the remote dimmers **504** may open their charge pulse detect window, which may be used by the remote dimmers **504** to stay in synchronization with the main dimmer **502**. For example, the rising edge of the charge pulse during the charging time period T_{CHRG} may be detected by the zero-cross detector **816** to establish the timing for the rest of the line cycle. The AD line **509** may be in a high impedance state (i.e., the high impedance state) during the charge pulse window T_{CPW} .

Although illustrated as comprising the charging time period T_{CHRG} and the communication time period T_{COMM} during the positive half-cycle of the AC voltage waveform **1000** but not the negative half-cycle of the AC voltage waveform **1000**, in one or more embodiments, the AC voltage waveform **1000** may include a charging time period T_{CHRG} and a communication time period T_{COMM} during the negative half-cycle of the AC voltage waveform **1000** but not the positive half-cycle of the AC voltage waveform **1000**.

FIG. 11 is a diagram of an example of a payload format for communication between the main dimmer **502** and the remote dimmer **504**, **514**. A packet **1100** may comprise two frames. The first frame may comprise a frame number **1102** and event data **1104**. The second frame may comprise a frame number **1106**, event type **1108**, device address **1110**, and an error detection **1112**. The frame number field **1102** and the frame number field **1106** may identify which frame of the packet **1100** is being sent. The frame number **1102** and the frame number **1106** may comprise one bit each. The event data **1104** may comprise the data being communicated between the main dimmer and the remote dimmer. The event data **1104** may comprise fifteen bits. The event type **1108** may indicate the type of packet **1100** communicated via the AD line **509**. For example, the event type **1108** may encode the possible packet types that will be communication via the AD line **509**. The event type **1108** may comprise seven bits. The device address **1110** may identify the source device of the packet **1100**. The device address **1110** may comprise three bits. The error detection field **1112** may encode the forward error detection result to be used by the receiving device to validate the packet **1100**. For example, the error detection field **1112** may comprise a multi-bit cyclic redundancy check (CRC) (e.g., a five bit CRC) that may be used by the receiving device to validate the packet **1100**.

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FIG. 12 is a flowchart of an example of a user interface procedure **1700** executed periodically by the control circuit **714** of the main dimmer **502**, e.g., once every 10 msec. The user interface procedure **1700** may selectively execute one of three routines depending upon the state of the main dimmer **502**. If the main dimmer **502** is in an "Idle" state (i.e., the user is not actuating the touch sensitive actuator **610**) at step **1710**, the control circuit **714** may execute an Idle routine **1800**. If the main dimmer **502** is in an "ActiveHold" state (i.e., the user is presently actuating the touch sensitive actuator **610**) at step **1720**, the control circuit **714** may execute an ActiveHold routine **1900**. If the main dimmer **502** is in a "Release" state (i.e., the user has recently ceased actuating the touch sensitive actuator **610**) at step **1730**, the control circuit **714** may execute a Release routine **2000**.

FIG. 13 is a flowchart of an example of the Idle routine **1800**, which may be executed periodically when the main dimmer **502** is in the Idle state. The control circuit **714** may change the state of the main dimmer **502** to the ActiveHold state when the user actuates the touch sensitive actuator **610**. For example, if there is activity on the touch sensitive actuator **610** of the main dimmer **502** at step **1810**, an activity counter may be incremented at step **1812**. Otherwise, the activity counter may be cleared at step **1814**. The activity counter may be used by the control circuit **714** to ensure that the main dimmer **502** changes to the ActiveHold state (e.g., only changes to the ActiveHold state) in response to an actuation of the touch sensitive actuator **610** and not as a result of noise or some other undesired impulse. The use of the activity counter may be similar to a software "debouncing" procedure for a mechanical switch. If the activity counter is not less than a maximum activity counter value $AMAX$ at step **1816**, then the state of the main dimmer **502** is set to the ActiveHold state at step **1818**. Otherwise, the Idle routine **1800** may exit.

FIG. 14 is a flowchart of an example of the ActiveHold routine **1900**, which may be executed once every half-cycle when the touch sensitive actuator **610** is being actuated, i.e., when the main dimmer **502** is in the ActiveHold state. The control circuit **714** may make a determination as to whether the user has stopped using, i.e., released, the touch sensitive actuator **610**. If there is no activity on the touch sensitive actuator **610** at step **1910**, the control circuit **714** may increment an "inactivity counter" at step **1912**. The control circuit **714** may use the inactivity counter to make sure that the user is not still actuating the touch sensitive actuator **610** before entering the Release mode. If the inactivity counter is less than a maximum inactivity counter value $IMAX$ at step **1914**, the ActiveHold routine **1900** may exit. Otherwise, the state of the main dimmer **502** may be set to the Release state at step **1915**, and the routine **1900** may exit.

If there is activity on the touch sensitive actuator **610** at step **1910**, the control circuit **714** may generate an audible sound at step **1916** using the audible sound generator **718**. An example of the generation of the audible sound is described in greater detail in commonly-assigned U.S. Pat. No. 7,608,948, issued Oct. 27, 2009, entitled TOUCH SCREEN WITH SENSORY FEEDBACK, the entire disclosure of which is hereby incorporated by reference. The control circuit **714** may determine where along the length of the actuation member **612** that the touch sensitive actuator is being actuated at step **1918**. If the touch sensitive actuator **610** is being actuated in the toggle area, i.e., the lower portion **612B** of the actuation member **612**, at step **1920**, the control circuit **714** may process the actuation of the touch sensitive actuator as a toggle. If the lighting load **508** is presently off at step **1922**, the control circuit **714** may turn

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the lighting load on. For example, the control circuit 714 may illuminate the lower portion 612B of the actuation member 612 white at step 1924 and dim the lighting load 508 up to the preset level, i.e., the desired lighting intensity of the lighting load, at step 1926. Further, the control circuit 714 may load a digital message into the TX buffer at step 1928. The message description of the digital message may comprise, for example, a light level command and the message data comprises the preset level.

If the lighting load is presently on at step 1922, the control circuit 714 illuminates the lower portion 612B of the actuation member 612 orange at step 1932 and controls the lighting load 508 to off at step 1934. At step 1928, the control circuit 714 loads a digital message into the TX buffer, where the message description is a light level command and the message data comprises zero percent (or off).

If the touch sensitive actuator 610 is not being actuated in the toggle area at step 1920, the upper portion 612A is being actuated and the location of the actuation on the touch sensitive actuator 610 is representative of the desired intensity level of the lighting load 508. At step 1936, the control circuit 714 may illuminate the upper portion 612A of the actuation member 612 appropriately, i.e., as a bar graph representative of the present intensity of the lighting load 508. The control circuit 714 may dim the lighting load 508 to the appropriate level as determined from the location of the actuation of the touch sensitive actuator 610 at step 1938. At step 1928, the control circuit 714 loads the TX buffer with a digital message having a light level command as the message description and the present intensity level as the message data.

FIG. 15 is a flowchart of an example of the Release routine 2000, which may be executed after the control circuit 714 sets the state of the dimmer state to the Release state at step 1915 of the ActiveHold routine 1900. The control circuit 714 may store the present intensity level of the lighting load 508 in the memory 718 at step 2010. At step 2012, the control circuit 714 may store one or more entries of the last digital message to be transmitted in response to the actuation of the touch sensitive actuator 610 into the TX buffer, for example, such that the main dimmer 502 may send one or more identical digital messages to the remote dimmers 504 to ensure that the remote dimmers received the digital message. The control circuit 714 may set the state of the main dimmer 502 to the Idle state at step 2014, and the Release routine 2000 may exit.

The message description of the digital messages transmitted between the main dimmer 502 and the remote dimmers 504 may comprise an advanced programming mode (APM) command, i.e., a command to adjust an advanced programming feature, such as a protected preset, a fade rate, and/or the like. If an advanced programming mode feature is modified at the main dimmer 502, the main dimmer 502 may transmit to the remote dimmers 504 a digital message having the message description containing the APM command and the message data comprising the APM feature to change and the value to change the APM feature to. For example, the digital message may be transmitted one or more times during the Release routine 2000. An example of an advanced programming mode is described in greater detail in commonly-assigned U.S. Pat. No. 7,190,125, issued Mar. 13, 2007, entitled PROGRAMMABLE WALLBOX DIMMER, the entire disclosure of which is hereby incorporated by reference.

FIG. 16 is a flowchart of an example of a RX buffer procedure 2100 executed periodically by the control circuit 714 of the main dimmer 502, e.g., once every positive or

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negative half-cycle. If there is a digital message in the RX buffer at step 2110, the control circuit 714 may determine whether the message description of the digital message contains an APM command at step 2112 or a light level command at step 2114. If the message description is an APM command at step 2112, the APM feature is modified in the memory 718 at step 2116 and the procedure 2100 exits. If the message description is a light level command at step 2116 and the message data of the digital message is zero percent (i.e., off) at step 2118, the control circuit 714 may illuminate the toggle area (i.e., the lower portion 612B of the actuation member 612) at step 2120, and/or may control the lighting load 508 to off at step 2122. On the other hand, if the message data for the light level command is an intensity greater than zero percent at step 2118, the control circuit 714 may illuminate the toggle area white at step 2124, and/or may illuminate the upper portion 612A of the actuation member 612 appropriately (i.e., as a bar graph representative of the present intensity of the lighting load 508) at step 2126. The control circuit 714 may control the intensity of the lighting load 508 to the appropriate level as determined from the message data of the digital message at step 2128 and the procedure 2100 may exit.

FIG. 17 is a flowchart of an example of a multi-location control procedure 2200 executed by the control circuit 714 of the main dimmers 502. The multi-location control procedure 2200 may be executed periodically, e.g., once every line cycle. The procedure 2200 may begin at step 2210 when the zero-crossing detector 716 signals a zero-crossing to the control circuit 714 (e.g., at the beginning of the charging time T_{CHRG} as shown in FIG. 10). Upon receiving the zero-crossing signal, the control circuit 714 may start the charging time T_{CHRG} at step 2212. During the charging time T_{CHRG} at 2214, the main dimmer 502 may charge the power supply 830 of the remote dimmer 504. For example, the control circuit 714 may render the transistors Q906 and Q908 conductive to charge the capacitor 940 of the remote dimmer 504.

At 2216, the control circuit 714 may determine if the charging time T_{CHRG} has ended. If not, then the control circuit 714 may continue to charge the power supply 830 of the remote dimmer 504. If the charging time T_{CHRG} has ended, the control circuit 714 may start a communication time T_{COMM} at 2218.

During the communication time T_{COMM} , the control circuit 714 may perform a communication routine at 2220. For example, the control circuit 714 may transmit a digital message to the remote dimmer 504 and/or receive a digital message from the remote dimmer 504 via control of the AD line 509 by the sender (i.e., placing the AD line 509 in the active pull-up state and/or the active-pull down state). To transmit a digital message, the control circuit 714 may render the transistor Q906 conductive. Then, to place the AD line 509 in the active pull up state to communicate a "1" bit, the control circuit 714 may render the transistor Q908 conductive and the transistor Q912 non-conductive. To place the AD line 509 in the active pull down state to communicate a "0" bit, the control circuit 714 may render the transistor Q908 non-conductive and the transistor Q912 conductive. Therefore, the control circuit 714 may inversely control the transistors Q908 and Q912 in a complementary manner to communicate a "1" bit or a "0" bit. To receive a digital message from the remote dimmer 504 during the communication time T_{COMM} , the control circuit 714 of the main dimmer 502 may render the transistor Q910 conductive. During the communication time T_{COMM} , the control circuit

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714 may render the transistor 910 conductive, such that the control circuit 714 is able to receive a digital message from the remote dimmer 504.

At 2222, the control circuit 714 may determine if the communication time T_{COMM} has ended. If not, the control circuit 714 may continue to perform the communication routine. If the communication time T_{COMM} has ended, the control circuit 714 may place the AD line 509 in a high impedance state at 2224, for example, until the next charging time period T_{CHRG} . For example, the control circuit 714 may render the transistor Q906 conductive and the transistors Q908, Q910, and Q912 non-conductive to place the AD line 509 in a high impedance state.

FIG. 18 is a flowchart of an example of a multi-location control procedure 2300 executed by the control circuit 814 of the remote dimmers 504, 514. The multi-location control procedure 2300 may be executed periodically, e.g., once every line cycle. The procedure 2300 may begin at step 2310 when the zero-crossing detector 816 signals a zero-crossing to the control circuit 814 (e.g., at the beginning of the charging time T_{CHRG} as shown in FIG. 10). Upon receiving the zero-crossing signal, the control circuit 814 may start the charging time T_{CHRG} at step 2312. During the charging time T_{CHRG} at 2314, the power supply 830 of the remote dimmer 504 may be charged by the main dimmer 502. For example, during the charging time T_{CHRG} , the control circuit 814 may render the transistors Q930, Q932, and Q934 non-conductive so that the capacitor 830 of the remote dimmer 504 may be charged.

At 2316, the control circuit 814 may determine if the charging time T_{CHRG} has ended. If not, then the control circuit 814 may continue to render the transistors Q930, Q932, and Q934 non-conductive so that the power supply 830 may be charged. If the charging time T_{CHRG} has ended, the control circuit 814 may start a communication time T_{COMM} at 2318.

During the communication time T_{COMM} , the control circuit 814 may perform a communication routine at 2320. For example, the control circuit 814 may transmit a digital message to the main dimmer 502 and/or receive a digital message from the main dimmer 502 via control of the AD line 509 by the sender (i.e., placing the AD line 509 in the active pull-up state and/or the active-pull down state). To receive a digital message, the control circuit 814 may render the transistor Q932 conductive. To transmit a digital message, the control circuit 814 may render the transistor Q934 conductive. Then, to place the AD line 509 in the active pull up state to communicate a "1" bit, the control circuit 814 may render the transistor Q934 conductive and the transistor Q930 non-conductive. To place the AD line 509 in the active pull down state to communicate a "0" bit, the control circuit 814 may render the transistor Q934 non-conductive and the transistor Q930 conductive. Therefore, the control circuit 814 of the remote dimmer 504 may inversely control the transistors Q934 and Q930 in a complementary manner to communicate a "1" bit or a "0" bit. During the communication time T_{COMM} , the control circuit 814 may render the transistor 932 conductive, such that the control circuit 814 is able to receive a digital message from the main dimmer 502.

At 2322, the control circuit 814 may determine if the communication time T_{COMM} has ended. If not, the control circuit 814 may continue to perform the communication routine. If the communication time T_{COMM} has ended, the control circuit 814 may place the AD line 509 in a high impedance state at 2324, for example, until the next charging time period T_{CHRG} . For example, the control circuit 814 may

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render the transistors Q930, Q932, and Q934 non-conductive to place the AD line 509 in a high impedance state.

At 2326, the control circuit 814 may determine if the window delay period TDELAY is complete. If the delay period TDELAY is complete, then the control circuit 814 may open the charge pulse window T_{CPW} at 2328. During the charge pulse window T_{CPW} , the control circuit 814 may monitor for a charge pulse that may occur during a subsequent charging time period T_{CHRG} during a subsequent line cycle. The detection of the charge pulse during the charge pulse window T_{CPW} may be used by the control circuit 814 to stay in synchronization with the main dimmer 502. For example, the rising edge of the charge pulse during the charging time period T_{CHRG} may be detected by the zero-cross detector 816 to establish the timing for the rest of the line cycle. As such, the control circuit 814 may start a subsequent charging time T_{CHRG} , e.g., return to 2312, upon detecting the charge pulse.

Since the digital messages transmitted between the main dimmers 502 and the remote dimmers 504 may include APM commands, the APM features of the load control system 500/510 may be modified using the user interface 600 of the main dimmer 502 and/or a remote dimmer 504. The main dimmer 502 and the remote dimmers 504 may be used to adjust local advanced programming features (i.e., of the main dimmer 502) and global advanced programming features (i.e., affecting the main dimmer 502 and one or more of the remote dimmers 504).

Although described with reference to a main dimmer and a remote dimmer, one or more embodiments described herein may be used with other load control devices. For example, one or more of the embodiments described herein may be performed by a variety of load control devices that are configured to control of a variety of electrical load types, such as, for example, a LED driver for driving an LED light source (e.g., an LED light engine); a screw-in luminaire including a dimmer circuit and an incandescent or halogen lamp; a screw-in luminaire including a ballast and a compact fluorescent lamp; a screw-in luminaire including an LED driver and an LED light source; a dimming circuit for controlling the intensity of an incandescent lamp, a halogen lamp, an electronic low-voltage lighting load, a magnetic low-voltage lighting load, or another type of lighting load; an electronic switch, controllable circuit breaker, or other switching device for turning electrical loads or appliances on and off; a plug-in load control device, controllable electrical receptacle, or controllable power strip for controlling one or more plug-in electrical loads (e.g., coffee pots, space heaters, other home appliances, and the like); a motor control unit for controlling a motor load (e.g., a ceiling fan or an exhaust fan); a drive unit for controlling a motorized window treatment or a projection screen; motorized interior or exterior shutters; a thermostat for a heating and/or cooling system; a temperature control device for controlling a heating, ventilation, and air conditioning (HVAC) system; an air conditioner; a compressor; an electric baseboard heater controller; a controllable damper; a humidity control unit; a dehumidifier; a water heater; a pool pump; a refrigerator; a freezer; a television or computer monitor; a power supply; an audio system or amplifier; a generator; an electric charger, such as an electric vehicle charger; and an alternative energy controller (e.g., a solar, wind, or thermal energy controller). A single control circuit may be coupled to and/or adapted to control multiple types of electrical loads in a load control system.

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What is claimed is:

1. An electric load control device comprising:
 user interface circuitry to receive user input indicative of
 a target electric load output parameter;
 power input circuitry to receive an alternating current (AC) supply from a controllably conductive device disposed in a remote electric load control device;
 accessory dimmer (AD) communication interface circuitry to bidirectionally exchange information with the remote electric load control device; and
 control circuitry to:
 receive, from operatively coupled zero detect circuitry, a first input indicative of a zero crossing of the AC supply;
 cause operatively coupled power supply circuitry to charge using the received AC supply for a first portion of an AC half cycle responsive to receipt of the first input;
 receive, via the user interface circuitry, the user input indicative of a first target electric load output parameter value for an electric load device operatively coupled to the controllably conductive device;
 responsive to receipt of the user input, generate a first message that includes data representative of the first target electric load output parameter value;
 provide a buffer period for a second portion of the AC half cycle after passage of the first portion of the AC half cycle; and
 cause a transmission, via the AD communication interface circuitry, of the generated first message during a third portion of the AC half cycle after the passage of the second portion of the AC half cycle.
2. The electric load control device of claim 1:
 wherein the user interface further comprises a display device; and
 wherein the control circuitry to further:
 cause the display device to generate an output indicative of the received first target electric load output parameter responsive to receipt of the user input.
3. The electric load control device of claim 2 wherein the control circuitry to further:
 receive, via the AD communication interface circuitry, a second message that includes data representative of a second target electric load output parameter value; and
 cause the display device to generate an output indicative of the received second target electric load output parameter value responsive to receipt of the second message.
4. The electric load control device of claim 1 wherein the control circuitry to further:
 monitor for a charge pulse window during a portion of the AC cycle prior to transition of the AC supply from a negative half cycle to a positive half cycle.
5. The electric load control device of claim 1 wherein the control circuitry to further, responsive to a placement of the controllably conductive device on the line side of the electric load device:
 charge the power supply for the first portion of a positive half cycle;
 provide the buffer period for the second portion of the positive half cycle; and
 cause the transmission, via the AD communication interface circuitry, of the generated first message during the third portion of the positive half cycle.

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6. The electric load control device of claim 1 wherein the control circuitry to further, responsive to a placement of the controllably conductive device on the load side of the electric load device:
 charge the power supply for the first portion of a negative half cycle;
 provide the buffer period for the second portion of the negative half cycle; and
 cause the transmission, via the AD communication interface circuitry, of the generated first message during the third portion of the negative half cycle.
7. An electric load control method, comprising:
 receiving, by control circuitry, a first input indicative of a zero crossing of the AC supply from operatively coupled zero detect circuitry;
 causing, by the control circuitry, operatively coupled power supply circuitry to charge using an alternating current (AC) supply for a first portion of a half cycle of the AC supply responsive to receipt of the first input;
 receiving, by the control circuitry, a user input indicative of a first target electric load output parameter value for an electric load device operatively coupled to a controllably conductive device disposed in a remote electric load controller;
 generating, by the control circuitry, a first message that includes data representative of the first target electric load output parameter value responsive to receipt of the user input;
 providing, by the control circuitry, a buffer period for a second portion of the half cycle of the AC supply after passage of the first portion of the half cycle of the AC supply; and
 causing, by the control circuitry, a transmission of the generated first message during a third portion of the half cycle of the AC supply after passage of the second portion of the half cycle of the AC supply via the AD communication interface circuitry.
8. The method of claim 7, further comprising:
 causing, by the control circuitry, a generation of a user-perceptible output indicative of the received first target electric load output parameter responsive to receipt of the user input.
9. The method of claim 8, further comprising:
 receiving, by the control circuitry, a second message that includes data representative of a second target electric load output parameter value; and
 causing, by the control circuitry, a generation of a user-perceptible output indicative of the received second target electric load output parameter value responsive to receipt of the second message.
10. The method of claim 7, further comprising:
 monitoring, by the control circuitry, for a charge pulse window during a portion of the AC cycle immediately prior to a transition of the AC supply from a negative half cycle to a positive half cycle.
11. The method of claim 7, further comprising, responsive to a placement of the controllably conductive device on the line side of the electric load device:
 causing, by the control circuitry, operatively coupled power supply circuitry to charge using the AC supply for a first portion of a positive half cycle of the AC supply;
 providing, by the control circuitry, the buffer period for the second portion of the positive half cycle of the AC supply and

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causing, by the control circuitry, the transmission of the generated first message during the third portion of the positive half cycle of the AC supply.

12. The method of claim 7, further comprising, responsive to a placement of the controllably conductive device on the line side of the electric load device:

causing, by the control circuitry, operatively coupled power supply circuitry to charge using the AC supply for a first portion of a negative half cycle of the AC supply;

providing, by the control circuitry, the buffer period for the second portion of the negative half cycle of the AC supply; and

causing, by the control circuitry, the transmission of the generated first message during the third portion of the negative half cycle of the AC supply.

13. A non-transitory, machine-readable, storage device that includes instructions that, when executed by an electric load controller, cause the electric load controller to:

receive a first input indicative of a zero crossing of the AC supply from operatively coupled zero detect circuitry;

cause operatively coupled power supply circuitry to charge using an alternating current (AC) supply for a first portion of a half cycle of the AC supply responsive to receipt of the first input;

receive a user input indicative of a first target electric load output parameter value for an electric load device operatively coupled to an controllably conductive device disposed in a remote electric load controller; generate a first message that includes data representative of the first target electric load output parameter value responsive to receipt of the user input;

provide a buffer period for a second portion of the half cycle of the AC supply after passage of the first portion of the half cycle of the AC supply; and

cause a transmission of the generated first message during a third portion of the half cycle of the AC supply after passage of the second portion of the half cycle of the AC supply via the AD communication interface circuitry.

14. The non-transitory, machine-readable, storage device of claim 13 wherein the instructions, when executed by the electric load controller, cause the electric load controller to: cause a generation of a user-perceptible output indicative of the received first target electric load output parameter responsive to receipt of the user input.

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15. The non-transitory, machine-readable, storage device of claim 14 wherein the instructions, when executed by the electric load controller, cause the electric load controller to:

receive a second message that includes data representative of a second target electric load output parameter value; and

cause a generation of a user-perceptible output indicative of the received second target electric load output parameter value responsive to receipt of the second message.

16. The non-transitory, machine-readable, storage device of claim 13 wherein the instructions, when executed by the electric load controller, cause the electric load controller to:

monitor for a charge pulse window during a portion of the AC cycle immediately prior to a transition of the AC supply from a negative half cycle to a positive half cycle.

17. The non-transitory, machine-readable, storage device of claim 13 wherein the instructions, when executed by the electric load controller, cause the electric load controller to, responsive to a placement of the controllably conductive device on the line side of the electric load device:

cause the operatively coupled power supply circuitry to charge using the AC supply for a first portion of a positive half cycle of the AC supply;

provide the buffer period for the second portion of the positive half cycle of the AC supply and

cause the transmission of the generated first message during the third portion of the positive half cycle of the AC supply.

18. The non-transitory, machine-readable, storage device of claim 13 wherein the instructions, when executed by the electric load controller, cause the electric load controller to, responsive to a placement of the controllably conductive device on the line side of the electric load device:

cause the operatively coupled power supply circuitry to charge using the AC supply for a first portion of a negative half cycle of the AC supply;

provide the buffer period for the second portion of the negative half cycle of the AC supply; and

cause the transmission of the generated first message during the third portion of the negative half cycle of the AC supply.

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